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Evolution of open quantum systems under continuous measurement of their environment

STUDENT:
Jules STEVENOT

SUPERVISOR:
Bruno BELLOMO[1]

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[1]Equipe ΦTh, Institut UTINAM - CNRS 6213, OSU THETA.

ABSTRACT

This Master's thesis investigates the dynamics of **open quantum systems subjected to continuous measurement of their environment**, within the framework of quantum trajectories. Starting from the **input-output formalism**, we derive the Lindblad master equation under the Born-Markov approximation, describing the unconditional (unmonitored) evolution of a system coupled to a continuum of bosonic environmental modes. We then consider the case where the environment is continuously monitored, focusing on two fundamental detection schemes: the **photo-detection** and the **homodyne detection**. For each scheme we derive the corresponding stochastic master equation governing the **conditional state** of the system. We show that photo-detection yields a jump-like unravelling driven by a Poisson process, while homodyne detection yields a diffusive unravelling driven by a Wiener process. We further discuss the linearization of these equations and their reformulation in terms of Kraus operators, which provides the bases for numerical simulation. These theoretical results are illustrated by numerical simulations, implemented in `Julia` using the `QuantumToolbox.jl` package. In particular, we simulate the dynamics of a qubit and of a harmonic oscillator coupled to a zero-temperature bath. For both detection schemes, individual quantum trajectories are simulated and averaged over a large number of realizations. The results confirm that **the average over trajectories reproduces the unconditional Lindblad dynamics**, consistent with the unravelling interpretation.

Ce mémoire de Master examine la dynamique des **systèmes quantiques ouverts soumis à une mesure continue de leur environnement**, inscrit dans le cadre des trajectoires quantiques. En partant du **formalisme entrée-sortie** (input-output), nous dérivons l'équation maîtresse de Lindblad sous l'approximation de Born-Markov, qui décrit l'évolution inconditionnelle (sans mesure continue) d'un système couplé à un continuum de modes environnementaux bosoniques. Nous considérons ensuite le cas où l'environnement est mesuré en continu, en nous concentrant sur deux schémas de détection fondamentaux : la **photodétection** et la **détection homodyne**. Pour chaque cas de mesure, nous dérivons l'équation stochastique maîtresse correspondante régissant l'**état conditionnel** du système. Nous montrons que la photodétection conduit à un "dénouement" (unravelling) par sauts, piloté par un processus de Poisson, tandis que la détection homodyne donne lieu à un dénouement diffusif, dirigé par un processus de Wiener. Nous présentons de surcroît la linéarisation de ces équations stochastiques maîtresses et leur reformulation en termes d'opérateurs de Kraus, ce qui fournit les bases de la simulation numérique. Ces résultats théoriques sont illustrés par des simulations numériques implémentées en `Julia` à l'aide du package `QuantumToolbox.jl`. En particulier, nous simulons la dynamique d'un qubit et d'un oscillateur harmonique couplés à un bain de température nulle. Pour les deux schémas de détection, les trajectoires quantiques individuelles sont simulées et moyennées sur un grand nombre de réalisations. Les résultats confirment que **la moyenne des trajectoires reproduit la dynamique inconditionnelle de Lindblad**, ce qui est cohérent avec l'interprétation par du dénouement des trajectoires.

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

CONTENTS

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, i.e., isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe the dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2]However, in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we consider the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them is the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a quantum system has a tremendous impact on its properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the measured observable — leading to a loss of the system's properties of interest. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states. To illustrate this property, we can think about the *Schrödinger's cat* thought experiment, which corresponds formally to a superposition of 'the cat is alive' and 'the cat is dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead. It is then "stuck" in the state measured and cannot be expressed as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering dramatically the state of the system, we extract pieces of information about it. The measurements can even be conducted continuously, maximizing the amount of data that an observer can obtain. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first chapter covers an academical progression from the basics of quantum mechanics up to the aforementioned theories. A second chapter will cover the photodetection and homodyne detection of the environment of open quantum systems. The final section of this second chapter presents simulation results for both the qubit and the harmonic oscillator, comparing the two detection methods on each of them.

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

Chapter I

Basics of quantum mechanics

I.1 Generalities on closed quantum systems

I.1.1 States and time evolution

In quantum mechanics, a physical system is defined by the relation $|\psi\rangle \in \mathcal{H}$, where $|\psi\rangle$ is the vector representing the system's state living in the Hilbert space $\mathcal{H}$. The Hamiltonian $\hat{H}$ is a specific Hermitian[2] operator corresponding to the classical quantity associated with the total energy of a system. This relationship will be further detailed in Section I.1.2. The evolution of a system is governed by

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle\,. \tag{I.1}$$

The solution of this Schrödinger equation can be expressed as an unitary[3] operator $\hat{U}$, such that

$$|\psi(t)\rangle = \hat{U}(t, t_0)\,|\psi(t_0)\rangle\,, \tag{I.2}$$

with the initial condition $U(t_0, t_0) = \mathbb{I}$. By injecting the previous relation in the state equation, we get

$$\frac{\mathrm{d}\hat{U}}{\mathrm{d}t}(t, t_0) = -\frac{i}{\hbar}\hat{H}(t)\hat{U}(t, t_0). \tag{I.3}$$

We will simply assume the Hamiltonian to be time-independent in the following, nevertheless, solutions exist in the time-dependent case [3]. For us, the solution of the Schrödinger equation is simply

$$|\psi(t)\rangle = \underbrace{e^{-\frac{i\hat{H}(t-t_0)}{\hbar}}}_{\hat{U}(t,t_0)}\,|\psi(t_0)\rangle\,. \tag{I.4}$$

This solution is indeed **unitary** and we can use $\hat{U}^\dagger(t, t_0)$ on the left of both sides of Eq. (I.4) to obtain $|\psi(t_0)\rangle = \hat{U}^\dagger(t, t_0)\,|\psi(t)\rangle$. This last property yields the **reversible** nature of the evolution according to the Schrödinger equation.

I.1.2 Observables

In quantum mechanics, any physical quantity is represented by an **observable** which is an operator fulfilling two properties: the eigenvectors of the observable form an orthonormal basis of the Hilbert space $\mathcal{H}$, and, the corresponding eigenvalues are necessarily real [2]. In finite-dimensional Hilbert spaces, the definition of Hermitian operator[4] is sufficient for it to be an observable. In the case of infinite dimension Hilbert spaces, the definition of an observable requires more rigorous mathematical foundations (self-adjoint

[2] $\hat{H}^\dagger = \hat{H}$, where † denotes the conjugate transpose, defining the Hermitian adjoint.

[3] $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{I}$, where $\mathbb{I}$ is the identity.

[4] The eigenvectors of a Hermitian operator always span the entire Hilbert space (spectral theorem). Using the Gram-Schmidt process, we can always find a complete orthonormal basis from the operator's eigenspaces.

operators, the details of which lie beyond the purview of this study). In this sense, we only consider finite dimension cases in the following. We thus define the interval $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document, also, we lighten the notation by omitting $\hat{\bullet}$ symbols on operators hereafter.

A natural operator whose eigenbasis $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ is particularly significant is the Hamiltonian H, as its eigenvalues $\{E_n\}_{n\in I_{\mathcal{H}}}$ correspond to the allowed energy values of a system. We can understand the relation between an operator, its eigenvectors and eigenvalues with its corresponding eigenequation

$$\forall n \in I_{\mathcal{H}}, \quad H\,|\phi_n\rangle = E_n\,|\phi_n\rangle\,. \tag{I.5}$$

Since $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ forms a basis of the Hilbert space, one can describe any state $|\psi\rangle$ in this very basis:

$$\forall\,|\psi\rangle \in \mathcal{H}, \qquad |\psi\rangle = \sum_{n\in I_{\mathcal{H}}} c_n\,|\phi_n\rangle\,, \tag{I.6}$$

where $c_n = \langle\phi_n|\psi\rangle$. This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian.

Finally, we mention the necessity of normalizing states, which is important when calculating probabilities (the Section I.2 makes it obvious). In order to normalize a state, one needs to set $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$, which is equivalent to the condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$.

I.1.3 Purity of states

The concept of *pure state* arises from the necessity to write the state of a system in a different way than that used so far in this document. Indeed, certain ways of preparing a system (such as a statistical mixture [2]) require another formalism because such processes prevent the state of the system from being expressed as a single $|\psi\rangle$ (such as a superposition of state vectors). To take into account such cases, we introduce the density matrix hereafter.

A density matrix ρ is an operator of the Hilbert space $\mathcal{H}$ (and also an operator of the dual space $\mathcal{H}^*$), which follows a certain set of properties.

- ρ is Hermitian: $\rho = \rho^\dagger$, implying it can be diagonalized and necessarily with a positive eigenvalues, $\mathrm{Sp}(\rho) \subset \mathbb{R}^+$.
- $\mathrm{Tr}\,[\rho] = 1$ necessarily.
- ρ is positive semi-definite: $\forall\,|\psi\rangle \in \mathcal{H}, \quad \langle\psi|\rho|\psi\rangle \geq 0$.

The density matrix formalism is particularly useful because it generalizes Dirac formalism. Indeed, if a state can be written as $|\psi\rangle$ then $\rho = |\psi\rangle\langle\psi|$ and we call it a **pure state**. Otherwise, the state is non-pure and can only be expressed with a density matrix. Also, pure states satisfy stronger properties than those defining a general density matrix. Specifically in the case of pure states, the density operator is idempotent by construction ($\rho^2 = \rho$), which implies that $\mathrm{Tr}\left[\rho^2\right] = \mathrm{Tr}\,[\rho] = 1$.

I.1.4 Postulates of quantum mechanics

Physicists have developed a core framework of quantum mechanics, the main properties of which are summarized here as the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathrm{End}(\mathcal{H})$ (an endomorphism of the Hilbert space), which respects the properties I.1.3. Pure states can be described with $|\psi\rangle \in \mathcal{H}$, or with the pure density matrix $\rho_p = |\psi\rangle\langle\psi|$.
2. Any physical quantity $\mathcal{A}$ is associated with an operator $\hat{A}$, called observable. In the context of finite dimension space, an observable is simply an Hermitian operator.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues $\{a_n\}_{n\in I_{\mathcal{H}}}$ of $\hat{A}$, defined by $\hat{A}\,|a_n\rangle = a_n\,|a_n\rangle$.
4. When measuring $\mathcal{A}$ on a normalized state, the probability to get one of the possible eigenvalues is $\mathbb{P}[a_n]$ (which exact expression depends on the nature of observable's spectrum).
5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement collapses into the corresponding eigenstate $|\phi_n\rangle$.
6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the *Schrödinger* equation:
$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = -\frac{i}{\hbar}H(t)\,|\psi(t)\rangle\,. \tag{I.7}$$
For density matrices, the evolution is driven by the *Liouville-von Neumann* equation:
$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\,[H(t),\rho(t)]\,, \tag{I.8}$$
where $[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is called the commutator of $\hat{A}$ and $\hat{B}$.

I.1.5 Multipartite systems

In quantum mechanics a system can be composed of multiple subsystems, we then call the global composition a multipartite system. The density matrix introduced in the previous sections is a useful tool to describe the state of such system and subsystems. If we consider a Hilbert space $\mathcal{H}_{\mathcal{AB}}$ composed of two physical systems A and B: $\mathcal{H}_{\mathcal{AB}} = \mathcal{H}_A \otimes \mathcal{H}_B$. Note that this implies $\dim(\mathcal{H}_{AB}) = \dim(\mathcal{H}_A) \times \dim(\mathcal{H}_B)$. The total system state is therefore represented by a density matrix $\rho_{AB} \in \mathrm{End}(\mathcal{H}_A \otimes \mathcal{H}_B)$. The expression of the density matrix representing the state of the subsystem A is defined by
$$\rho_A = \mathrm{Tr}_B[\rho_{AB}], \tag{I.9}$$
and is called a *reduced density matrix*. Moreover, we defined in this equation a new function called a *partial trace*. The idea behind the notation is to get rid of the B components in the tensor product of the total state ρ_{AB}. Explicitly, we can express the mean value of an operator of the subsystem A in the total space. Let $\hat{O} = \hat{O}_A \otimes \mathbb{I}_B$, and specify the bases of A and B: namely $\{|\phi_a\rangle\}_a$ and $\{|\phi_b\rangle\}_b$. The average of the operator $\hat{O}$ is expressed as
$$\begin{aligned}\left\langle \hat{O} \right\rangle = \mathrm{Tr}\left[\rho_{AB}\hat{O}\right] &= \sum_{a=1}^{\dim(\mathcal{H}_A)}\sum_{b=1}^{\dim(\mathcal{H}_B)} \left((\langle\phi_a|\otimes\langle\phi_b|)\rho_{AB}\left(\hat{O}_A\otimes\mathbb{I}_B\right)(|\phi_b\rangle\otimes|\phi_a\rangle)\right) \\ &= \sum_{a=1}^{\dim(\mathcal{H}_A)} \langle\phi_a| \underbrace{\left(\sum_{b=1}^{\dim(\mathcal{H}_B)} \langle\phi_b|\,\rho_{AB}\,|\phi_b\rangle\right)}_{\mathrm{Tr}_B[\rho_{AB}]:=\rho_A} \hat{O}_A\,|\phi_a\rangle = \mathrm{Tr}_A\left[\rho_A\hat{O}_A\right].\end{aligned} \tag{I.10}$$
Hence, the expectation value of any local observable acting on subsystem A is completely determined by the reduced density matrix ρ_A. Remarkably, evaluating this local expectation value does not require full knowledge of the global state ρ_{AB}, since in the end we have that $\left\langle \hat{O} \right\rangle = \mathrm{Tr}_A\left[\rho_A\hat{O}_A\right]$. Nevertheless, because ρ_A is obtained by tracing out the subsystem B, it naturally retains and accounts for all quantum and classical correlations that existed between the two subsystems.

The bipartite results shown above can be generalized to multipartite systems. However, in the following we will consider a system S coupled to an environnement E only, which is thus bipartite.

I.2 Quantum Measurement Theory (QMT)

In the QMT [6, 7], measuring consists in applying a projector to a state, so, we call this process a *projective measurement*. The set of projective measurements is called a Projection-Valued Measure (PVM) [8]. For the sake of simplicity, we work — as until now — with non-degenerate spectra for the observables. Since an observable has a set of eigenvectors $\{|a_n\rangle\}_{n\in I_{\mathcal{H}}}$, we can build

$$P_{a_n} = |a_n\rangle\langle a_n| \,. \tag{I.11}$$

This definition implies four important properties:

$$P_{a_n}^\dagger = P_{a_n}, \qquad P_{a_n}^2 = P_{a_n}, \qquad P_{a_n}P_{a_m} = P_{a_n}\delta_{nm}, \qquad \sum_n P_{a_n} = \mathbb{I}.$$

To correctly represent the concept of a quantum measurement, the projective measurement needs to fulfill some requirements. The first requirement is to generate the probability of observing the result of a measurement on a system. Thus, the probability to observe a_n is defined by

$$\mathbb{P}_{|\psi\rangle}[a_n] = \langle\psi|P_{a_n}|\psi\rangle = |\langle\psi|a_n\rangle|^2, \qquad \mathbb{P}_\rho[a_n] = \mathrm{Tr}\,[\rho P_{a_n}]\,, \tag{I.12}$$

where the subscripts explicit which formalism is used to represent the state (a vector or a density matrix). We recall that the vector representation of a state is valid only in the case of pure states. The expression of the state after the measurement becomes

$$\overline{|\psi_{a_n}\rangle} = P_{a_n}|\psi\rangle\,, \qquad \overline{\rho_{a_n}} = P_{a_n}\rho P_{a_n}. \tag{I.13}$$

Here the $\bar{\bullet}$ symbol means that the state is not normalized, which will be an important notation for later, see Section II.5. By normalizing we find the proper expression of a state after a measurement with result a_n:

$$|\psi_{a_n}\rangle = \frac{P_{a_n}|\psi\rangle}{\sqrt{\langle\psi|P_{a_n}|\psi\rangle}}, \qquad \rho_{a_n} = \frac{P_{a_n}\rho P_{a_n}}{\mathrm{Tr}\,[\rho P_{a_n}]}. \tag{I.14}$$

The obtained state is said **conditioned by the measurement** result a_n, hence the importance of the index a_n on states obtained after measurement. We point out that the normalization factor is exactly the probability to observe the result a_n for the density matrix (or the square root of the probability, when using ket formalism). In the end, projective measurements fulfill the properties of quantum mechanics by construction because they generate probabilities (respecting the fact they need to be positive and their sum is 1), and can also be used to write the expression of collapsed states after measurement.

However, the definition of PVM has been found too restrictive to represent every kind of measurement allowed by quantum mechanics [9, 10]. One can define a more general kind of measurement, that fulfills quantum mechanics needs while relaxing some properties: the orthogonality between each measure elements is not required to generate probabilities. Therefore, one can build the so-called Positive Operator-Valued Measure (POVM), implicitly defined by the probability to observe the result a_n:

$$\mathbb{P}_\psi[a_n] = \langle\psi|\Pi_{a_n}|\psi\rangle\,, \qquad \mathbb{P}_\rho[a_n] = \mathrm{Tr}\,[\rho\Pi_{a_n}]\,, \tag{I.15}$$

where the operator Π_{a_n} is called the *effect* [7] for the result a. Moreover, the set of all effects $\{\Pi_{a_n}\}_{n\in I_{\mathcal{H}}}$ forms the POVM. As stated, the only properties we force the POVM to respect are:

$$\text{positive semi-definite: } \forall\,|\phi\rangle\in\mathcal{H},\ \langle\phi|\Pi_{a_n}|\phi\rangle\geq 0, \qquad \sum_{n\in I_{\mathcal{H}}}\Pi_{a_n} = \mathbb{I}.$$

These properties imply that POVM generate proper probabilities. Furthermore, the (normalized) states obtained after finding the result a_n are now

$$|\psi_{a_n}\rangle = \frac{K_{a_n}|\psi\rangle}{\sqrt{\langle\psi|\Pi_{a_n}|\psi\rangle}}, \qquad \rho_{a_n} = \frac{K_{a_n}\rho K_{a_n}^\dagger}{\mathrm{Tr}\,[\rho\Pi_{a_n}]}, \tag{I.16}$$

where we introduced the so-called *Kraus* operator K_{a_n}, which respects $\Pi_{a_n} = K_{a_n}^\dagger K_{a_n}$.

POVM can describe processes of quantum measurement where the results are ignored, just as in the thought experiment of a statistical mixture of states developed in Appendix **??**. In such a case, we can describe statistically the state without knowing the measurement results:

$$\rho = \sum_{n\in I_{\mathcal{H}}}\mathbb{P}[a_n]\rho_{a_n} = \sum_{n\in I_{\mathcal{H}}}K_{a_n}\rho K_{a_n}^\dagger. \tag{I.17}$$

The corresponding relationship for PVM is obtained by replacing both K_{a_n} and $K_{a_n}^\dagger$ by P_{a_n}.

Finally, one can observe that POVM generalize PVM in the sense that projectors are special cases of POVM elements (adding self-adjoint property and orthogonality to POVM gives a PVM). Furthermore, PVMs make states collapse on eigenstates that are necessarily orthogonal to each other, which is not necessarily the case for POVMs. A brief example of POVM forcing a state to collapse onto "non-orthogonal states" is when considering the state $|\psi\rangle = \frac{|g\rangle+|e\rangle}{\sqrt{2}} \equiv |+\rangle$. If we choose the Kraus operators $K_0 = |g\rangle\langle g|$ and $K_1 = |+\rangle\langle e|$, which fulfill $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$, and, that form the POVM $\{|g\rangle\langle g|\,, |e\rangle\langle e|\}$. Then, when the result of the measurement is 0, we find that the state collapses to $|\psi_0\rangle = |g\rangle$, and when finding 1 it collapses to $|\psi_1\rangle = |+\rangle$. However, $\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}} \neq 0$, thus the states on which the state can collapse after measurement are not orthogonal to each other.

I.3 Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. However, this description applies to the ideal case where the quantum system is considered isolated from any external interaction. Instead, in the OQS theory systems interact locally with an environment, which itself interacts with a wider environment, etc... To avoid considering infinite chains of interaction, the theory truncates the environment to a local vicinity of the system. Thus, we approximate the total system T as a closed quantum system, within which the quantum system of interest S is called an open quantum system. Moreover, the coupling between the system S and its environment E induces a dynamical regime that differs fundamentally from that of closed quantum systems.

I.3.1 Building the formalism

We will present here a derivation of the Lindblad master equation using the *input-output formalism* [5, 11]. This derivation can alternatively be carried out using a microscopic approach [3]. Since a full derivation of the input-output formalism is mathematically challenging, we will only provide its key equations and their physical meaning.

As seen in Section I.1.5, the Hilbert space of the total system is divided into $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. We consider as initial state a factorized state: $\rho_T = \rho_S \otimes \rho_E$. From now on we will use a typographic convention where $\rho_T \equiv \boldsymbol{\varrho}$ and $\rho_S \equiv \rho$, such that the last assumption rewrites $\boldsymbol{\varrho} = \rho \otimes \rho_E$. The Hamiltonian H_S must be time independent and depends on which physical system is studied. In some cases, we can use time dependant Hamiltonian by using a changing of frame (rotating frame transformation), leading to a time independent Hamiltonian in this frame. However, we will consider the simplest time independent case in this work. Instead, the environment is chosen as a collection of quantum harmonic oscillators, the environment Hamiltonian being thus defined by

$$H_E = \int_0^{+\infty} \hbar\omega b_\omega^\dagger b_\omega \mathrm{d}\omega, \tag{I.18}$$

where the operators of the environment satisfy $\left[b_\omega, b_{\omega'}^\dagger\right] = \delta(\omega-\omega')$. This Hamiltonian represents a continuum of bosonic modes of frequency $\omega \in \mathbb{R}^+$, where each mode is represented by a quantum harmonic oscillator. We define the so-called interaction Hamiltonian by

$$H_I = i\hbar \int_0^{+\infty} \sqrt{\frac{k(\omega)}{2\pi}} \left(c b_\omega^\dagger - c^\dagger b_\omega\right) \mathrm{d}\omega, \tag{I.19}$$

where c is an operator of the system, and $\omega \mapsto k(\omega)$ is a function representing the coupling strength between S and E for each considered mode at frequency ω. This coupling is treated as a perturbation: the interaction has a smaller impact on the dynamics than the free evolution. This form of interaction Hamiltonian can be found, for example, in the case of a two-level system interacting with a radiation field via a dipole interaction, once the rotating wave approximation has been performed. Consequently by using Eq. (I.18,I.19), we define the total Hamiltonian:

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{I.20}$$

By switching to the interaction picture (detailed in Appendix A), we can get rid of the free dynamics governed by H_0. Indeed, in the interaction picture the Liouville-von Neumann equation becomes

$$\frac{\mathrm{d}\tilde{\boldsymbol{\varrho}}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I, \tilde{\boldsymbol{\varrho}}\right], \tag{I.21}$$

in which we used the following notation: for any operator $O(t)$ we denote by $\tilde{O}(t)$ its expression in the interaction picture, such that $\tilde{O}(t) = U_0^\dagger(t,t_0)O(t)U_0(t,t_0)$. The unitary operator U_0 is the free evolution operator generated by H_0, and is defined as $U_0(t,t_0) = e^{-\frac{i}{\hbar}H_0(t-t_0)}$.

Before calculating explicitly the expression of $\tilde{H}_I$, we need to make some assumptions which are listed below.

- For the *first Markovian assumption* we consider the interaction strength to be constant in a (large) bandwidth $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$ (ω_0 is the natural frequency of S), so that $\forall\omega \in \mathcal{W}, \quad k(\omega) = k$. The coupling strength is considered negligible outside of $\mathcal{W}$.
- The system operator acquires a time-dependent phase in the interaction picture: $\tilde{c}(t) = ce^{-i\omega_0(t-t_0)}$. This ansatz is motivated by the expression of the system's Hamiltonian in the interaction picture when considering elementary systems [5] (two-levels quantum system or harmonic oscillator for example). The ansatz is equivalent to $[c, H_S] \propto c$.
- We assume that $k \ll \omega_0$, which establishes $\tau = \frac{1}{k}$ as the dominant timescale, dictating the slow non-unitary dynamics of the system.

We can now express the interaction Hamiltonian in the interaction picture by

$$\tilde{H}_I(t) = i\hbar\sqrt{k}\left[cb_t^\dagger - c^\dagger b_t\right], \tag{I.22}$$

in which we defined the input field operator b_t as

$$b_t = \frac{1}{\sqrt{2\pi}}\int_{\mathcal{W}} b_\omega e^{-i(\omega-\omega_0)(t-t_0)}\mathrm{d}\omega. \tag{I.23}$$

To characterize the environmental operators b_t, we can compute the canonical commutation relation

$$\left[b_t, b_{t'}^\dagger\right] = \frac{1}{\sqrt{2\pi}}\int_{-\theta}^{\theta} e^{-i\omega(t-t')}\mathrm{d}\omega = \delta(t-t'). \tag{I.24}$$

To obtain Eq. (I.24), we assumed the frequency bandwidth $\mathcal{W}$ to be much larger than all other frequencies of the dynamics, implying in particular $k \ll \theta$. In this limit, the integration in Eq. (I.24) can be extended to the entire real line [5, 11, 12], which yields the definition of the Dirac delta distribution. Note that by defining $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$, we implicitly state that $\omega_0 \geq \theta$. This condition gives again the third assumption mentioned earlier: $k \ll \theta \leq \omega_0 \Rightarrow k \ll \omega_0$. The delta-distribution we just obtained in Eq. (I.24) is not satisfying because the commutator becomes singular at equal times. We will now open a brief parenthesis on input operators and Wiener increments to bypass this singularity.

We define a Wiener process in quantum stochastic calculus as

$$W(t,t_0) = \int_{t_0}^{t} b_{t'}\mathrm{d}t'. \tag{I.25}$$

We can further derive a differential quantity using calculus [13], called a quantum Wiener increment and defined as

$$\mathrm{d}W_t = \lim_{\delta t \to \mathrm{d}t}\left[W(t+\delta t, t_0) - W(t,t_0)\right] = b_t\mathrm{d}t. \tag{I.26}$$

Using Eq. (I.25), the quantum Wiener increment canonical commutation relation (at same times) is

$$\left[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\right] = \mathrm{d}t. \tag{I.27}$$

However, by calculating directly the same commutator with the right hand side of Eq. (I.26), we find

$$\left[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\right] = \left[b_t, b_t^\dagger\right]\mathrm{d}t^2. \tag{I.28}$$

When we combine this result with Eq. (I.27), we get

$$\left[b_t, b_t^\dagger\right]\mathrm{d}t = \mathbb{I}. \tag{I.29}$$

If we now define a set of bosonic operators B_t such that $B_t = b_t\sqrt{\mathrm{d}t}$, then

$$\left[B_t, B_t^\dagger\right] = \mathbb{I}, \tag{I.30}$$

which is the canonical commutation relation of creation/annihilation bosonic operators[5]. Since $\{B_t\}_{t\in\mathbb{R}}$ forms an infinite set of proper annihilation operators, the environment can be modeled as a continuous collection of quantum harmonic oscillators. In this framework, at any time t, the system couples to the corresponding temporal bosonic mode defined by B_t. After evolving over the infinitesimal duration $\mathrm{d}t$, the system decouples from the mode at t and interacts with the next mode at $t + \mathrm{d}t$.

We now make use of one more assumption to derive the evolution of the state $\rho(t)$ of the open quantum system S.

- The *Born-Markov approximation*: at each time t, the total state $\boldsymbol{\varrho}(t)$ is a factorized state with the "same" reduced state for the environment,

$$\forall t \in \mathbb{R}, \quad \tilde{\boldsymbol{\varrho}}(t) = \tilde{\rho}(t) \otimes \nu_t, \tag{I.31}$$

where $\nu_t = |0\rangle\langle 0|_t$ is the vacuum state of the bosonic operator B_t. Coupling with the corresponding vacuum state (at time t) implies that the average of the input modes canonical anticommutator is

$$\left\langle \left\{ b_t, b_{t'}^\dagger \right\} \right\rangle = \delta(t - t'), \tag{I.32}$$

where by definition the anticommutator of A and B is $\{A, B\} = AB + BA$. This equation and Eq. (I.24) form the Markovian assumptions of this framework [5]: the bosonic mode B_t is completely uncorrelated with the other modes at different times due to the delta-correlation of b_t. The global state evolves over an infinitesimal time $\mathrm{d}t$ as

$$\tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t) = U_I(t + \mathrm{d}t, t)\, \overbrace{[\tilde{\rho}(t) \otimes \nu_t]}^{\tilde{\boldsymbol{\varrho}}(t)}\, U_I^\dagger(t + \mathrm{d}t, t), \tag{I.33}$$

where $U_I(t + \mathrm{d}t, t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)dt}$, as explained in Appendix A.

I.3.2 The Lindblad master equation

We just calculated the evolution of the state $\boldsymbol{\varrho}$ of the total system over an infinitesimal time $\mathrm{d}t$. In order to obtain the evolved state $\rho(t + \mathrm{d}t)$ of the system S, we want to "trace out" the environment (as explained in Section I.1.5) by applying the following partial trace

$$\tilde{\rho}(t + dt) = \mathrm{Tr}_E[\tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t)]. \tag{I.34}$$

To write this expression into explicit terms, we develop the evolved total state and thus, we need to explicit the expression of $U_I(t + \mathrm{d}t, t)$. To do so, we can rewrite it using the environmental ladder operators, using Eq. (I.22), as

$$U_I(t + \mathrm{d}t, t) = \exp\left[\sqrt{k\mathrm{d}t}\left(c \otimes B_t^\dagger - c^\dagger \otimes B_t\right)\right]. \tag{I.35}$$

[5]In the input-output formalism, the annihilation operator acts on its Fock basis as $B_t |n\rangle_t = \sqrt{n}\,|n-1\rangle_t$ (with the vacuum condition $B_t |0\rangle_t = 0$), whereas the adjoint (creation operator) acts as $B_t^\dagger |n\rangle_t = \sqrt{n+1}\,|n+1\rangle_t$.

The subtlety here is that the unitary evolution operator scales with $\sqrt{\mathrm{d}t}$, and because $\mathrm{d}t = o(\sqrt{dt})$ expanding to order $\mathrm{d}t$ is already a fine description of the dynamics. Consequently, we expand the exponential to second order so that we capture all terms up to $\mathrm{d}t$, giving

$$U_I(t+\mathrm{d}t,t) \approx \mathbb{I}\otimes\mathbb{I} + \left(c\otimes B_t^\dagger - c^\dagger\otimes B_t\right)\sqrt{k\mathrm{d}t} + \left(c^{\dagger 2}\otimes B_t^2 + c^2\otimes B_t^{\dagger 2} - cc^\dagger\otimes B_t^\dagger B_t - c^\dagger c\otimes B_t B_t^\dagger\right)\frac{k\mathrm{d}t}{2}. \quad \text{(I.36)}$$

We can now rewrite the evolution of $\tilde{\boldsymbol{\varrho}}$ during the infinitesimal time $\mathrm{d}t$, given by Eq. (I.33), using the previous expansion:

$$\tilde{\boldsymbol{\varrho}}(t+\mathrm{d}t) \approx [\tilde\rho(t)\otimes|0\rangle\langle 0|_t]\times 1 + \left[\tilde\rho(t)c^\dagger\otimes|0\rangle\langle 1|_t + \text{h.c.}\right]\times\sqrt{k\mathrm{d}t} + \left[\frac{\sqrt{2}}{2}\tilde\rho(t)c^{\dagger 2}\otimes|0\rangle\langle 2|_t + \text{h.c.} - \frac{1}{2}\tilde\rho(t)cc^\dagger\otimes|0\rangle\langle 0|_t + \text{h.c.} + c\tilde\rho(t)c^\dagger\otimes|1\rangle\langle 1|_t\right]\times k\mathrm{d}t, \quad \text{(I.37)}$$

where "h.c." stands for hermitian conjugate of the last term. We can now apply the partial trace Tr_E on the last equation. The evolved state of the system S is thus expressed as

$$\tilde\rho(t+\mathrm{d}t) = \tilde\rho(t) + k\left[c\tilde\rho(t)c^\dagger - \frac{c^\dagger c\tilde\rho(t)+\tilde\rho(t)c^\dagger c}{2}\right]\mathrm{d}t.$$

If we now form the difference quotient $\frac{\tilde\rho(t+\mathrm{d}t)-\tilde\rho(t)}{dt}$, in the infinitesimal limit, we find the differential equation followed by the state $\tilde\rho$ at all time,

$$\frac{\mathrm{d}\tilde\rho}{\mathrm{d}t}(t) = k\mathfrak{D}[c]\tilde\rho(t), \quad \text{(I.38)}$$

where we defined the super-operator $\mathfrak{D}[\hat O]\bullet \equiv \hat O\bullet\hat O^\dagger - \frac{\{\hat O^\dagger\hat O,\bullet\}}{2}$, called the dissipator. Finally, by coming back into the Schrödinger representation, we find the so-called Lindblad master equation:

$$\boxed{\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}[H_S,\rho(t)] + k\mathfrak{D}[c]\rho(t)}. \quad \text{(I.39)}$$

The proof is straightforward, however one should pay attention to the compensation of complex exponential when calculating $U_0cU_0^\dagger$, using the ansatz $\tilde c(t) = ce^{-i\omega_0(t-t_0)}$. Note that since ρ and c are operators of $\mathcal{H}_S$, any commutator with H_0 simplifies into a commutator with H_S, since trivially $\left[\hat O_S, H_E\right] = 0$ for any operator $\hat O_S$ of the system S.

Chapter II

Continuously measured open quantum systems

In the previous chapter, we went through the steps leading to the Lindblad master equation using the input-output formalism. In this derivation, the environment has been traced out to recover information on the system S from the global coupled system, leaving an impact on the dynamics of S, crystallized in the expression of the dissipator $\mathfrak{D}$. In the following we will focus on the impact of continuously monitoring the environment on the dynamics of the system S. Indeed, in the case of continuous measurement of the environment, the system state cannot follow the Lindblad master equation, and we expect to find a differential equation that takes into account the impact of the measurement. However, we will see in the following that any differential equation obtained from continuously monitoring the environment is strongly linked to the Lindblad master equation.

In Chapter I, we have developed the input-output formalism in which the system S interacts sequentially in time with harmonic oscillators. Indeed, at each time the environment is represented by an harmonic oscillator represented by the annihilation operator B_t. This operator combines all environmental frequency modes by construction, as defined in Eq. (I.23). A fundamental property we have explained is that these operators are mutually uncorrelated. Consequently, the environment is "refreshed" after each evolution, ensuring no memory effects or temporal dependence of environmental operators. Therefore, initially the system S couples to B_t at time t, then the total system evolves according to the Lindblad master equation over the period of time $\mathrm{d}t$. Immediately after this evolution, at time $t + \mathrm{d}t$, the system uncouples from B_t to couple with a fresh $B_{t+\mathrm{d}t}$, evolving again, and so forth. In the following, we will go one step further by measuring the environment after each coupled evolution step, as schematized in Figure 1.

This chapter will start by a brief presentation of the state of the art. After, two kinds of measurement will be presented: the photo-detection and the homodyne detection. After presenting these two processes, we will tackle the notion of linear quantum trajectories and also the Kraus operators formalism. Finally, we will present a section of numerical simulations to illustrate the different measurement schemes.

II.1 State of the art

This section aims to place the present work within the broader context of modern quantum physics. In the first chapter, we have introduced the fundamental concepts of Quantum Measurement Theory (QMT) and Open Quantum Systems (OQS), which provide the theoretical framework required to describe continuous quantum measurements.

QMT links the quantum formalism to classical measurement outcomes. Early quantum mechanics was developed without a complete measurement theory, but Heisenberg [14], Dirac [15], von Neumann [16] and Lüders [17] established the foundations of projective measurements. For a long time, these ideas were mainly of foundational interest, as experiments typically involved ensembles or destructive measurements. This framework was later extended to non-ideal measurements by Davies [18] and Kraus [19], providing a general description of quantum measurement processes. In parallel, continuously monitored quantum systems were investigated, notably through quantum jump models [20]. A major shift occurred with experiments

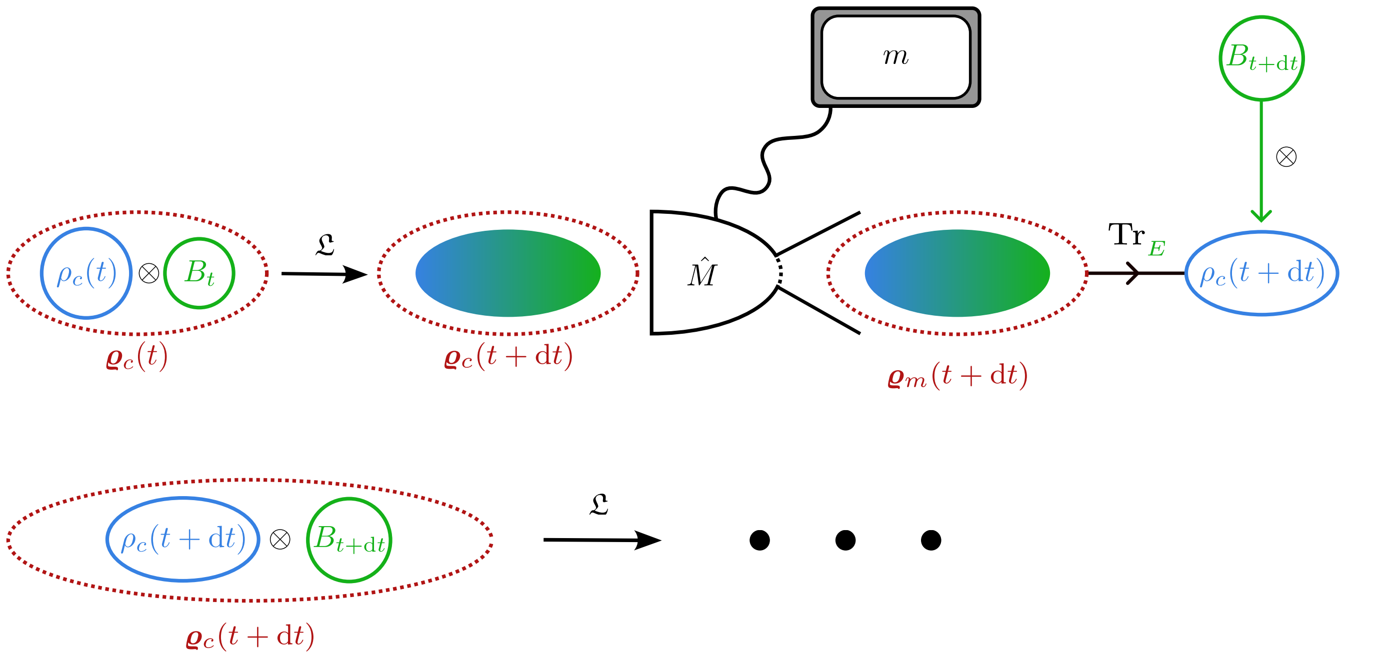


Figure 1: Scheme of continuous monitoring. In the first place the total system evolves from time t to $t+\mathrm{d}t$ according to the Lindblad master equation (denoted as $\mathfrak{L}$). We then measure the observable $\hat{M}$ (acting solely on the environment) on the total state and we denote the outcome as m. Then, by tracing out the environment, we find $\rho_c(t+\mathrm{d}t)$. To form the next total system $\varrho_c(t+\mathrm{d}t)$, the system couples instantaneously with the next temporal mode $B_{t+\mathrm{d}t}$, and so on. The process is repeated until the end of the monitoring. Note that before applying the partial trace, the system state is in general entangled with the environmental state due to the dynamics, this entanglement is represented by a gradient of color on the scheme. Moreover, it is important to realize that the index c at $t+\mathrm{d}t$ has been updated and includes the series of measurements up to time $t+\mathrm{d}t$ (and thus the result m obtained at time t in particular).

on single quantum systems, especially the observation of quantum jumps in trapped ions [21, 22, 23]. This required a dynamical description of measurement-conditioned quantum states, leading to stochastic evolution equations and the quantum trajectory formalism [24, 25, 26]. Independently, Belavkin developed quantum filtering theory by extending classical control concepts to continuously measured quantum systems [27, 28], establishing a rigorous framework for quantum trajectories in open systems. These developments naturally led to quantum feedback control, where measurement results are used in real time to steer dynamics [29, 30, 31, 32].

Beyond its fundamental interest, continuous quantum measurement is closely connected to several active research areas, including quantum parameter estimation [33, 34, 35] and quantum feedback control [36, 37]. These topics rely on the information extracted from measurement records and play a central role in modern applications. The relationships between these different research directions are illustrated in Fig. 2, extracted from the preface of *Wiseman and Milburn*'s Quantum Measurement and Control [38]. The diagram highlights how quantum measurement theory and open quantum systems provide the foundations for quantum trajectories, quantum feedback control, and their applications to quantum information processing.

II.2 Unconditional dynamics

The evolution of unmonitored open quantum systems is described by the Lindblad master equation, as seen in Section I.3. However, performing a measurement after each infinitesimal step makes the evolution dependent on each outcome, thereby altering the system's trajectory. If we consider the state of the system at time t, the state is conditional because of all the measurement outcomes from t_0 to t and is therefore noted as $\rho_c(t)$. The sequence of conditional states followed by a system is therefore called *a quantum trajectory* because the set of measurement outcomes gives a specific dynamical trajectory.

It is possible to rewrite the Lindblad master equation to yield the conditional states inside this uncon-

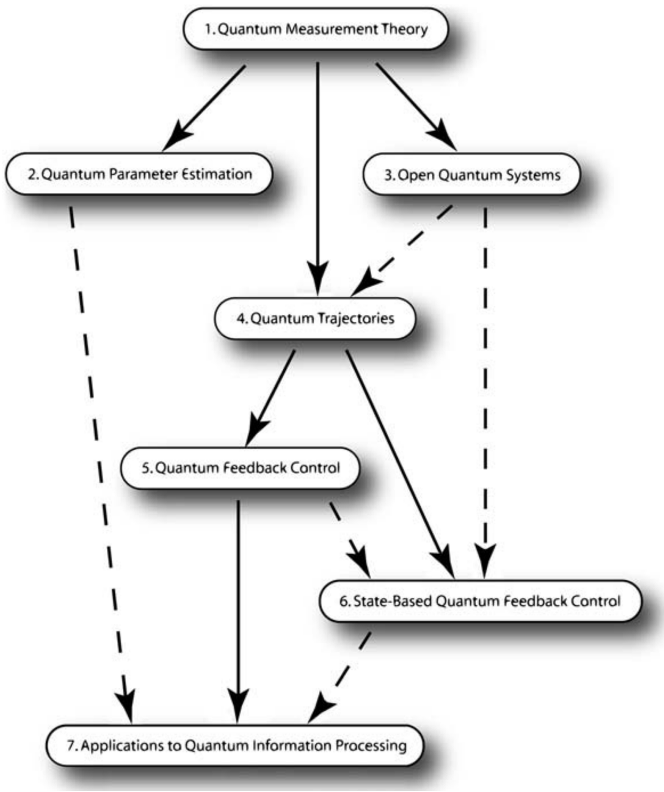


Figure 2: Conceptual map of the main topics. Quantum Measurement Theory and Open Quantum Systems form the basis of quantum trajectories, which in turn connect to quantum feedback control and applications in quantum information processing. Quantum parameter estimation and state-based feedback control are closely related research directions. Extracted from the preface of *Wiseman and Milburn*'s Quantum Measurement and Control [38].

ditional equation [39]. The rewriting proposed by *Carmichael* shows that the Lindblad master equation is decomposable into quantum trajectories. Therefore, we refer to quantum trajectories as *an unravelling of the source dynamics* because it is one trajectory among the many paths coming from the same equation. In this sense, when describing photo-detection or homodyne detection, we unravel the master equation of a specific measurement. Nevertheless, **on average, they recover the same Lindblad master equation, i.e., the unconditional dynamics**. We will show this result in the case of photo-detection and homodyne detection below.

Note that in order to lighten notations, we stop specifying the interaction picture by $\tilde{\bullet}$ in the following. However, when a change of representation occurs it will be specified. Furthermore, we work from now on in natural units where $\hbar = 1$.

II.3 Photo-detection

In the case where the environment is continuously monitored by a photo-detector, the mathematical impact is a projection of the environment on the corresponding Fock basis $\{|m\rangle\langle m|_t\}_{m\in\mathbb{N}}$. Indeed, photo-detection is a way to measure the number operator $B_t^\dagger B_t$ because the Fock basis is the eigenbasis of this operator. At first glance the measurement would only give information about the environnement. However, by performing a partial trace Tr_E over the total evolved state, we can extract information on the system. Without the evolution of the couple system-environment, the partial trace on the evolved total state would not give any information on the system whatsoever.

In the following, we find the expression of the conditional state obtained at time $t+\mathrm{d}t$ in the case of photo-detection. Let $\{|m\rangle\langle m|_t\}_m$ be a PVM (see Section I.2), so that the measurement projects the environment at time t on the PVM element $|m\rangle\langle m|_t$. According to Eq. (I.33), the total evolved system is expressed as

$$\boldsymbol{\varrho}_c(t+\mathrm{d}t) = U_I(t+\mathrm{d}t,t)\overbrace{\left[\rho_c(t)\otimes|0\rangle\langle 0|_t\right]}^{\boldsymbol{\varrho}_c(t)}U_I^\dagger(t+\mathrm{d}t,t), \tag{II.1}$$

where the subscript c indicates that the system state is conditional. If the condition is obvious, we choose not to write this subscript explicitly, instead we write for example ρ_m if the outcome is m. Note that our convention of notation about conditional states means that at time $t+\mathrm{d}t$, we denote the state as $\rho_c(t)$ instead of $\rho_m(t)$; the symbol c encodes the list of all measurement outcomes. We observe that the state in Eq. (II.1) is indeed conditional because we consider the measurement already present in the evolution before time t. After the projection on $|m\rangle\langle m|_t$ and after applying the partial trace Tr_E, we find

$$\rho_m(t+\mathrm{d}t) = \frac{\langle m|_t\,\boldsymbol{\varrho}_c(t+\mathrm{d}t)\,|m\rangle_t}{p_j(t+\mathrm{d}t)}, \tag{II.2}$$

where $p_m(t+\mathrm{d}t) = \mathrm{Tr}\left[\boldsymbol{\varrho}_c(t+\mathrm{d}t)(\mathbb{I}\otimes|m\rangle\langle m|_t)\right]$. Eq. (I.37) shows that only projections on $|0\rangle\langle 0|_t$ and $|1\rangle\langle 1|_t$ are relevant to compute $\rho_m(t+\mathrm{d}t)$, as shown in the following.

Measurement result 0

From the expansion in Eq. (I.37), we find

$$\rho_0(t+\mathrm{d}t) = \frac{\langle 0|_t\,\boldsymbol{\varrho}_c(t+\mathrm{d}t)\,|0\rangle_t}{p_0(t+\mathrm{d}t)} \approx \frac{\rho_c(t) - \left\{\rho_c(t), c^\dagger c\right\}\frac{k\mathrm{d}t}{2}}{p_0(t+\mathrm{d}t)}. \tag{II.3}$$

The normalization is obtained using the same expansion, and it gives

$$p_0(t+\mathrm{d}t) \approx 1 - \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t, \tag{II.4}$$

where we have defined the average labelled by t as $\left\langle \hat{O}\right\rangle_t = \mathrm{Tr}\left[\rho_c(t)\hat{O}\right]$. In the infinitesimal limit, we expand the denominator in Eq. (II.2), obtaining

$$\rho_0(t+\mathrm{d}t) \approx \bar{\rho}_0(t+\mathrm{d}t)\times\left(1 - \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t\right), \tag{II.5}$$

where we used the same notation as in Section I.2, $\bar{\rho}$ indicating an unnormalized state (here in the interaction picture). In particular, $\bar{\rho}_0(t+\mathrm{d}t)$ is the numerator of $\rho_0(t+\mathrm{d}t)$. Finally, by introducing the measurement super-operator $\mathfrak{H}[\hat{O}]\bullet = \hat{O}\bullet + \bullet\hat{O}^\dagger - \left\langle \hat{O}+\hat{O}^\dagger\right\rangle_t\bullet$, we can write at first order in $\mathrm{d}t$

$$\rho_0(t+\mathrm{d}t) = \rho_c(t) - \frac{k\mathrm{d}t}{2}\mathfrak{H}\left[c^\dagger c\right]\rho_c(t). \tag{II.6}$$

Measurement result 1

For this outcome, the same calculation method than the one used before still applies, and the conditional (unnormalized) state is expressed as

$$\bar{\rho}_1(t) = \langle 1|_t\,\boldsymbol{\varrho}(t+dt)\,|1\rangle_t \approx c\rho_c(t)c^\dagger k\mathrm{d}t. \tag{II.7}$$

The normalization is thus

$$p_1(t+\mathrm{d}t) = \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t. \tag{II.8}$$

Combined, the two above equations give the normalized conditional state

$$\rho_1(t+\mathrm{d}t) = \frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t}. \tag{II.9}$$

Mixing the results

From a statistical perspective, the measurement outcome at time $t+\mathrm{d}t$ is a Bernoulli-like process, yielding either 0 with probability $p_0(t+\mathrm{d}t)$ or 1 with probability $p_1(t+\mathrm{d}t)$. The full set of results up to time T is captured by the "sum" of results obtained between times t_0 and T. If for example we discretize time allowing to measure the system 30 times and that the sum of results is 20, it is obvious that it is composed of 10×0 outcomes and 20×1 outcomes. In this sense, we define the random variable $N(T,t_0) = \int_{t_0}^{T} \mathrm{d}N_t$, where in an infinitesimal slice of time $\mathrm{d}t$, the increment $\mathrm{d}N_t = N(t+\mathrm{d}t, t_0) - N(t, t_0)$ is either 0 or 1. This last observation leads to the fundamental relation: $\mathrm{d}N_t^2 = \mathrm{d}N_t$. In effect, $N(T,t_0)$ is a Poisson process and $\mathrm{d}N_t$ is a Poisson increment[7]. With the probabilities found above, we can calculate the stochastic average of $\mathrm{d}N_t$ to get

$$\mathbb{E}[\mathrm{d}N_t] = 0 \times p_0(t+\mathrm{d}t) + 1 \times p_1(t+\mathrm{d}t) = \left\langle c^\dagger c \right\rangle_t k\mathrm{d}t. \tag{II.10}$$

Note that the expression of $\mathbb{E}[\mathrm{d}N_t]$ is proportional to the expectation value $\langle c^\dagger c\rangle_t$. If we consider a system where c is the annihilation operator, this proportionality means that we can experimentally access to the mean occupation number. Therefore, the Poisson increment allows to write the differential conditional state $\mathrm{d}\rho_c(t+\mathrm{d}t)$ with respect to each possible outcome. In particular, $\mathrm{d}N_t = 0$ means that the differential conditional state is $\mathrm{d}\rho_0(t) = \rho_0(t+\mathrm{d}t) - \rho_c(t)$, while $\mathrm{d}N_t = 1$ means that $\mathrm{d}\rho_1(t) = \rho_1(t+\mathrm{d}t) - \rho_c(t)$. Consequently,

$$\mathrm{d}\rho_c(t) = [\rho_0(t+\mathrm{d}t) - \rho_c(t)](1 - \mathrm{d}N_t) + [\rho_1(t+\mathrm{d}t) - \rho_c(t)]\mathrm{d}N_t. \tag{II.11}$$

We can inject the results of Eqs. (II.6,II.9) in the last equation and by going back in the Schrödinger picture, we obtain

$$\boxed{\mathrm{d}\rho_c = -i[H_S, \rho_c]\mathrm{d}t - \frac{k}{2}\mathfrak{H}\left[c^\dagger c\right]\rho_c\mathrm{d}t + \left(\frac{c\rho_c c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c\right)\mathrm{d}N_t}. \tag{II.12}$$

This equation is called the photo-detection Stochastic Master Equation (SME), and we note that we went back to the Schrödinger picture to obtain this expression. Moreover, we have discarded terms of order $\mathrm{d}N_t\mathrm{d}t$ when calculating the expression of this SME. Indeed, we have shown that $\mathbb{E}[\mathrm{d}N_t] \propto \mathrm{d}t$, so $\mathbb{E}[\mathrm{d}N_t\mathrm{d}t] = (\mathrm{d}t)^2$, also since we can show that $\mathbb{V}[\mathrm{d}N_t] \propto \mathrm{d}t + o(\mathrm{d}t)$, it implies that $\mathbb{V}[\mathrm{d}N_t\mathrm{d}t] \propto (\mathrm{d}t)^2[\mathrm{d}t + o(\mathrm{d}t)] \propto \mathrm{d}t^3 + o(\mathrm{d}t^3)$. Thus, statistical terms as $\mathrm{d}N_t\mathrm{d}t$ are negligible compared to $\mathrm{d}t$ in average, and Appendix B details why we can also neglect these terms locally. Finally, when derived for a pure state the expression simplifies to the Stochastic Schrödinger Equation (SSE)

$$\mathrm{d}\left|\psi_c\right\rangle = \left[-iH_S\mathrm{d}t + \frac{k}{2}\left(\left\langle c^\dagger c\right\rangle_t - c^\dagger c\right)\mathrm{d}t + \left(\frac{c}{\sqrt{\langle c^\dagger c\rangle_t}} - 1\right)\mathrm{d}N_t\right]\left|\psi_c\right\rangle, \tag{II.13}$$

which conserves the purity of the state during the stochastic evolution. Note that we can derive the SME from the SSE using Itō calculus rule [5]:

$$\mathrm{d}\rho_c = \mathrm{d}\left|\psi_c\right\rangle\left\langle\psi_c\right| + \left|\psi_c\right\rangle\mathrm{d}\left\langle\psi_c\right| + \mathrm{d}\left|\psi_c\right\rangle\mathrm{d}\left\langle\psi_c\right|. \tag{II.14}$$

The importance of the SME and of the SSE is double because these equations also contain in average the **unconditional** dynamics, as shown in the following.

We will now show briefly that we can write $\mathbb{E}[\rho_c] = \rho$, where ρ is the unconditional state following the Lindblad master equation. If we average the photo-detection SME over all possible trajectories we find

$$\begin{aligned}\mathbb{E}[\mathrm{d}\rho_c](t) = -i[H_S, R(t)]\mathrm{d}t - \frac{k}{2}\Big\{c^\dagger cR(t) + R(t)c^\dagger c - 2\mathbb{E}\Big[\left\langle c^\dagger c\right\rangle_t \rho_c(t)\Big]\Big\}\mathrm{d}t \\ + \mathbb{E}\left\{\left[\frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c(t)\right]\mathrm{d}N_t\right\},\end{aligned} \tag{II.15}$$

[7]A Poisson increment follows a Poisson distribution with rate $\Lambda_t \in \mathbb{R}$: $\forall n \in \mathbb{N}$, $\mathbb{P}[\mathrm{d}N_t = n] = \frac{\Lambda_t^n}{n!}e^{-\Lambda_t}$. However, since we only have two possible outcomes (0 or 1), the rate is forced to be $\Lambda_t = 0$ because the sum of probabilities gives 1. By setting $\Lambda_t = \lambda_t\mathrm{d}t \to 0$ in the infinitesimal limit, we can find that $\mathbb{P}[\mathrm{d}N_t = 0] \approx 1 - \lambda_t\mathrm{d}t \to 1$ and $\mathbb{P}[\mathrm{d}N_t = 1] \approx \lambda_t\mathrm{d}t \to 0$. It is therefore trivial to show that $\mathbb{E}[\mathrm{d}N_t] = \lambda_t\mathrm{d}t$, and also that $\mathbb{V}[\mathrm{d}N_t] = \lambda_t\mathrm{d}t + \lambda_t^2\mathrm{d}t^2$.

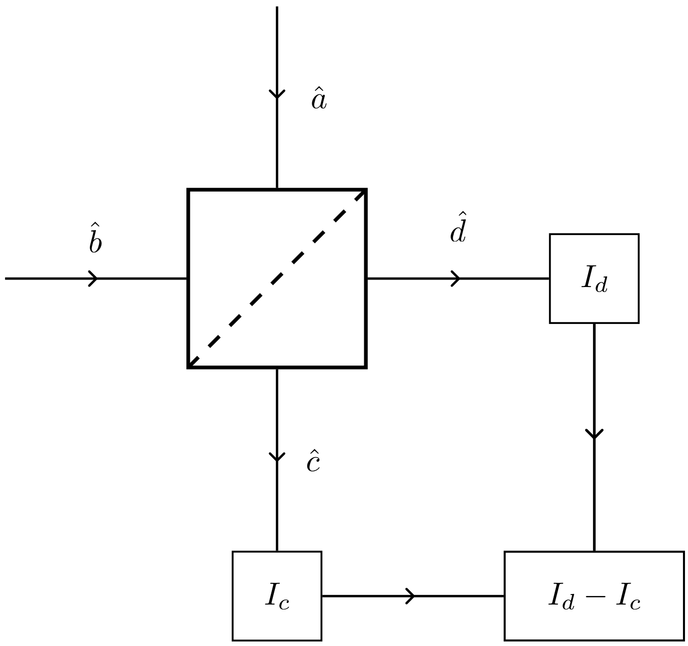


Figure 3: Schematic of the balanced homodyne detection method. The input $\hat{a}$ comes from the total system $S+E$ while $\hat{b}$ is a strong coherent field. The boxes represent photo-detectors measuring the fields after they passed through a beam-splitter (the box with a diagonal at the center). The apparatus calculates the difference of the photo-currents to obtain a (continuous) signal proportional to the quadrature of the system's field. We detail briefly this relationship in Appendix C.

where $R(t)=\mathbb{E}[\rho_c(t)]$. To explicit the last term, we need the relation $\mathbb{E}[f(\rho_c)(t)\mathrm{d}N_t]=k\mathbb{E}\left[\left\langle c^\dagger c\right\rangle_t f(\rho_c)(t)\right]\mathrm{d}t$, where f is a generic function [38]. So we obtain for the last term

$$\begin{aligned}\mathbb{E}\left\{\left[\frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t}-\rho_c(t)\right]\mathrm{d}N_t\right\}&=k\mathbb{E}\left\{\left\langle c^\dagger c\right\rangle_t\left[\frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t}-\rho_c(t)\right]\right\}\mathrm{d}t\\&=kcR(t)c^\dagger\mathrm{d}t-k\mathbb{E}\left[\left\langle c^\dagger c\right\rangle_t\rho_c(t)\right]\mathrm{d}t.\end{aligned}\quad\text{(II.16)}$$

By injecting this result in Eq. (II.15), we find

$$\mathbb{E}[\mathrm{d}\rho_c](t)=-i[H_S,R(t)]\mathrm{d}t+k\mathfrak{D}[c]R(t)\mathrm{d}t,\quad\text{(II.17)}$$

which is exactly the Lindblad master equation for the state $R(t)$. We know that this equation is the expression of an unconditional state evolution, thus we can state that $\mathbb{E}[\rho_c(t)]=\rho(t)$. Therefore for photo-detection, we have shown that averaging over all the quantum trajectories reforms the unconditional Lindblad master equation, and that the average over all possibilities of the conditional state is the unconditional state. This property is also valid for homodyne detection [7, 40, 41] and will be used without proper demonstration in the following section.

II.4 Homodyne detection

In the previous section, we derived the SME in the case of photo-detection and showed its link to the Lindblad master equation. While photo-detection monitors the number of excitations, the homodyne detection targets field quadratures. This process involves mixing the signal with a strong local oscillator before measurement [7]. By measuring the global output signal, we have access to the quadrature of the physical system of interest [42]. We depict the homodyne detection scheme in Figure 3.

This kind of measurement corresponds to projecting the environmental oscillator associated with B_t on the eigenvectors of the quadrature operator $\hat{x}_\theta$. This operator is defined by $\hat{x}_\theta=(e^{-i\theta}B_t+e^{i\theta}B_t^\dagger)/\sqrt{2}$, where $\theta\in\mathbb{R}$, and verifies $\left[\hat{x}_\theta,\hat{x}_{\theta+\frac{\pi}{2}}\right]=i$. Even if it is not possible to construct normalized eigenstates of this operator, we can build the states $|x_\theta\rangle$ satisfying $\hat{x}_\theta\,|x_\theta\rangle=x_\theta\,|x_\theta\rangle$ in such a way that $\langle x_\theta|x_\lambda\rangle=\delta(\theta-\lambda)$, and verifying the closure relation $\int_{\mathbb{R}} |x_\theta\rangle\langle x_\theta| \, \mathrm{d}x_\theta = 1$. We will use the following overlapping property [43] between these states and the Fock states,

$$\langle x_\theta|n\rangle_t = \frac{e^{-\frac{x_\theta}{2}} H_n(x) e^{-in\theta}}{\sqrt{2^n n!}\pi^{\frac{1}{4}}}, \tag{II.18}$$

where $H_n(x)$ is the (physical) Hermite polynomial of order n. In particular, we will later use

$$\langle x_\theta|0\rangle_t = \pi^{-\frac{1}{4}} e^{-\frac{x_\theta^2}{2}}, \qquad \langle x_\theta|1\rangle_t = \sqrt{2} x_\theta e^{-i\theta} \langle x_\theta|0\rangle_t . \tag{II.19}$$

Note that since the state of the environnement at time t is the corresponding vacuum state, the probability to find the outcome x_θ at time t (before evolving) is

$$p_{x_\theta}(t) = |\langle x_\theta|0\rangle_t|^2 = \frac{e^{-x_\theta^2}}{\sqrt{\pi}}. \tag{II.20}$$

Continuous homodyne measurement corresponds by definition to the projection of the environment at each time on the quadrature states, giving the conditional (unnormalized) state

$$\begin{aligned}\bar{\rho}_c(t+\mathrm{d}t) &= \mathrm{Tr}_E\Big[U_I(t+\mathrm{d}t,t)\boldsymbol{\varrho}_c(t)U_I^\dagger(t+\mathrm{d}t,t)(\mathbb{I}\otimes|x_\theta\rangle\langle x_\theta|)\Big] \\ &= \langle x_\theta|U_I(t+\mathrm{d}t,t)[\rho_c(t)\otimes|0\rangle\langle 0|_t]U_I^\dagger(t+\mathrm{d}t,t)|x_\theta\rangle ,\end{aligned} \tag{II.21}$$

with the corresponding normalization (probability)

$$p_{x_\theta}(t+\mathrm{d}t) = \mathrm{Tr}_S[\bar{\rho}_c(t+\mathrm{d}t)]. \tag{II.22}$$

We will now use the expansion of Eq. (I.37) to develop Eq. (II.21). However, we truncate it to $o\left(\sqrt{dt}\right)$ because $\mathrm{d}t$ is negligible compared to $\sqrt{\mathrm{d}t}$, giving for the conditional unnormalized evolved state

$$\bar{\rho}_c(t+\mathrm{d}t) = |\langle x_\theta|0\rangle_t|^2\rho_c(t) + \Big[c\rho_c(t)\langle x_\theta|1\rangle_t\langle 0|x_\theta\rangle + \rho_c(t)c^\dagger\langle x_\theta|0\rangle_t\langle 1|x_\theta\rangle\Big]\sqrt{k\mathrm{d}t} + o\left(\sqrt{\mathrm{d}t}\right). \tag{II.23}$$

Simplifying this expression and factorizing it, using Eqs. (II.19,II.20), we find

$$\bar{\rho}_c(t+\mathrm{d}t) = p_{x_\theta}(t)\Big\{\rho_c(t) + x_\theta\sqrt{2k\mathrm{d}t}\Big[c\rho_c(t)e^{-i\theta} + \rho_c(t)c^\dagger e^{i\theta}\Big]\Big\} + o\left(\sqrt{dt}\right), \tag{II.24}$$

with the normalization

$$p_{x_\theta}(t+\mathrm{d}t) = p_{x_\theta}(t)\Big\{1 + x_\theta\sqrt{2k\mathrm{d}t}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t\Big\} + o\left(\sqrt{\mathrm{d}t}\right) \tag{II.25}$$

$$\approx \frac{1}{\sqrt{\pi}} \times \exp\left\{-\left[x_\theta - \sqrt{\frac{k\mathrm{d}t}{2}}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t\right]^2\right\}. \tag{II.26}$$

We can deduce from Eq. (II.26) that the probability to find the result x_θ at time $t+\mathrm{d}t$ corresponds to a Gaussian distribution[8] (up to order $\sqrt{\mathrm{d}t}$). In this sense, the outcome of measurement is a random variable such that $x_\theta \hookrightarrow \mathcal{N}\left(\sqrt{\frac{k\mathrm{d}t}{2}}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t, \frac{1}{2}\right)$. By defining the random variable $\mathrm{d}y_t = \sqrt{2\mathrm{d}t}x_\theta$, we find $\mathbb{E}[\mathrm{d}y_t] = \sqrt{k}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t\mathrm{d}t$ and $\mathbb{V}[\mathrm{d}y_t] = \mathrm{d}t$. Consequently, we can statistically express this random variable as

$$\mathrm{d}y_t = \sqrt{k}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t\mathrm{d}t + \mathrm{d}w_t, \tag{II.27}$$

where $\mathrm{d}w_t$ follows a Gaussian distribution characterized by $\mathbb{E}[\mathrm{d}w_t] = 0$ and $\mathbb{V}[\mathrm{d}w_t] = \mathrm{d}t$. Such a random variable is called a classical Wiener increment and obeys the Itō rule $\mathrm{d}w_t^2 = dt$. This rule should be understood in the distributional sense as $\int_0^t \mathrm{d}w_{t'}^2 = \int_0^t \mathrm{d}t'$ (valid for any interval $[0,t]$) [40]. Experimentally,

[8] A Gaussian or normal distribution of mean μ and variance σ^2 is characterized by the following probability density function: $X \mapsto \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(X-\mu)^2}{2\sigma^2}\right]$. We denote by $X \hookrightarrow \mathcal{N}(\mu,\sigma^2)$ the fact that the random variable X follows a normal distribution with mean μ and variance σ^2.

we obtain at time t a photocurrent $I(t)$ from the apparatus of homodyne detection process. To make this photocurrent appear, we define the integrated result of homodyne detection at time T as

$$y_T = \int_0^T \mathrm{d}y_t = \int_0^T I(t)\mathrm{d}t. \tag{II.28}$$

Using Eq. (II.27), we can define the photocurrent in the distributional sense as

$$I(t) = \sqrt{k}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t + \xi(t), \tag{II.29}$$

where $\xi(t) = \frac{\mathrm{d}w_t}{\mathrm{d}t}$ is a classical white noise such that $\mathbb{E}[\xi(t)\xi(t')] = \delta(t-t')$ [5]. The definition of $\xi(t)$ implies $\mathbb{E}[\xi(t)] = 0$ and $\mathbb{V}[\xi(t)] = \frac{1}{\mathrm{d}t} \to +\infty$ (due to the Itō rule). However, when integrated inside of Eq. (II.28), the variance becomes a finite quantity. Moreover, the notation seems to define ξ as the derivative of w_t which is wrong because a Wiener process is not differentiable anywhere. Indeed, we can see it informally by forming the rate of change $\frac{W_{t+\mathrm{d}t}-W_t}{\mathrm{d}t} = \frac{\mathrm{d}w_t}{\mathrm{d}t} \sim \frac{1}{\sqrt{\mathrm{d}t}} \underset{\mathrm{d}t\to 0}{\longrightarrow} +\infty$. The correct usage of ξ is really as an integrand multiplied by $\mathrm{d}t$, as in Eq. (II.28). **Note that, up to a factor $1/\sqrt{2}$, the photocurrent $I(t)$ contains the system's average quadrature.** Indeed, the average of the quadrature operator for the system S is defined as

$$\langle c_\theta\rangle_t = \left\langle \frac{ce^{-i\theta} + c^\dagger e^{i\theta}}{\sqrt{2}}\right\rangle_t. \tag{II.30}$$

To find the homodyne SME, we inject Eq. (II.27) into Eqs. (II.24,II.25). By dividing these new equations, we form the normalized evolved state. We can then expand the denominator and find

$$\mathrm{d}\rho_c(t) = \sqrt{k}\mathfrak{H}\left[ce^{-i\theta}\right]\rho_c(t)\mathrm{d}w_t + o(\sqrt{\mathrm{d}t}). \tag{II.31}$$

The terms of order $o(\mathrm{d}t)$ are inferred as we know that the average over all trajectories must recover the Lindblad master equation, as explained in the last paragraph of Section II.3. However, the average of Eq. (II.31) trivially gives 0, so the correct SME (in the Schrödinger picture) is

$$\boxed{\mathrm{d}\rho_c = -i[H_S,\rho_c]\mathrm{d}t + k\mathfrak{D}[c]\rho_c\mathrm{d}t + \sqrt{k}\mathfrak{H}\left[ce^{-i\theta}\right]\rho_c\mathrm{d}w_t}, \tag{II.32}$$

where we recall that $\mathrm{d}w_t = \mathrm{d}y_t - \sqrt{k}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t \mathrm{d}t$.

If we consider the previous case of measurement, the photo-detection, we observe that the unravelling obeys to a jump-like process. Indeed, the integral $N(T) = \int_0^T \mathrm{d}N_t$ consists in a sum of quantities jumping from 0 to 1. Instead, the homodyne unravelling is following a diffusive process because $W(T) = \int_0^T \mathrm{d}w_t$ is a Gaussian white noise with variance T, thus linearly increasing with time. In this sense, the same Lindblad master equation can be unravelled via two different physical process (jump-like and diffusive), resulting in two different SMEs (photo-detection and homodyne detection respectively).

II.5 Linear quantum trajectories

The previously obtained SMEs are non-linear in ρ_c, implying heavy computational work in order to simulate a trajectory. Indeed, if we observe the two SMEs obtained in the previous sections, we can remark their non-linearity with respect to $\rho_c(t)$. The photo-detection SME, i.e., Eq. (II.12), is non-linear because of the term $\left\langle c^\dagger c\right\rangle_t = \mathrm{Tr}\left[c^\dagger c\rho_c(t)\right]$ at the denominator in the term multiplying $\mathrm{d}N_t$, and also because it is multiplied by $\rho_c(t)$ in the expression of $\mathfrak{H}\left[c^\dagger c\right]\rho_c(t)$. Similarly, in Eq. (II.32), the homodyne SME is non-linear because the term $\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t = \mathrm{Tr}\left[(ce^{-i\theta} + c^\dagger e^{i\theta})\rho_c(t)\right]$ is squared when we multiply $\mathfrak{H}[ce^{-i\theta}]\rho_c(t)$ by $\mathrm{d}w_t$. However, linear SMEs can be built at the cost of not using normalized states as we explain in the following.

To find the homodyne linear master equation, we replace $\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t$ by $\mu \in \mathbb{R}$ in Eq. (II.32). In particular, the replaced term is also involved in the normalization of the state at time $t+\mathrm{d}t$, as shown in Eq. (II.26). Therefore, the linear equation describes the evolution of non-normalized states, denoted as

$\breve{\rho}_c(t,\mu)$. We distinguish these non-normalized states from the ones introduced in Section I.2 noted $\bar{\rho}$ because their trace is not equal to the probability to find this trajectory, as it will be made clear further below. If we observe the effect of changing $\left\langle ce^{-i\theta}+c^\dagger e^{i\theta}\right\rangle_t$ into a constant, we realize that the homodyne outcome becomes $\mathrm{d}y_t=\sqrt{k}\mu\mathrm{d}t+\mathrm{d}w_t$. Therefore, a natural choice is to set $\mu=0$, yielding $\mathrm{d}y_t=\mathrm{d}w_t$. Consequently, the linear SME is expressed as

$$\mathrm{d}\breve{\rho}_c=-i[H_S,\breve{\rho}_c]\mathrm{d}t+k\mathfrak{D}[c]\breve{\rho}_c\mathrm{d}t+\sqrt{k}\Big(ce^{-i\theta}\breve{\rho}_c+\breve{\rho}_c c^\dagger e^{i\theta}\Big)\mathrm{d}y_t, \tag{II.33}$$

where we chose to simplify the notation $\breve{\rho}_c(t,\mu=0)=\breve{\rho}_c(t)$. This SME allows one to calculate iteratively $\breve{\rho}_c(t+\mathrm{d}t)=\breve{\rho}_c(t)+\mathrm{d}\breve{\rho}_c(t)$ for each timestep by simply generating a classical white noise $\mathrm{d}w_t$. Then, by dividing $\breve{\rho}_c(t+\mathrm{d}t)$ by its trace, one can find the expected (normalized) state $\rho_c(t+\mathrm{d}t)$.

In fact, building the linear SME introduces a different probability law than the one truly followed by the outcomes and states in the normalized case. Indeed, the outcome $\mathrm{d}y_t$ has to follow a different probability distribution when it is simply equal $\mathrm{d}w_t$ (as in the case of the linear SME). We denote as p_{true} the distribution which gives the relationship $\mathbb{E}_{p_{\text{true}}}[\rho_c(t)]=\rho(t)$, and as p_{ost} the distribution which gives $\mathbb{E}_{p_{\text{ost}}}[\breve{\rho}_c(t)]=\rho(t)$ ("ost" stands for ostensible [38]). The relationship between these two probabilities is

$$p_{\text{true}}(\text{traj.})=p_{\text{ost}}(\text{traj.})\times\mathrm{Tr}\left[\breve{\rho}_c(t)\right], \tag{II.34}$$

where "traj." stands for the quantum trajectory whose state at time t is $\breve{\rho}_c(t)$. This last equation shows how the trace of the solution of the linear SME encodes the (true) probability of the conditional state $\rho_c(t)$. If we explicit the expectation values according to the distributions p_{true} and p_{ost}, one can show that Eq. (II.34) can be used to prove[9] that $\mathbb{E}_{p_{\text{true}}}[\rho_c(t)]=\mathbb{E}_{p_{\text{ost}}}[\breve{\rho}_c(t)]$. Moreover, we have shown in the previous sections that $\mathbb{E}_{p_{\text{true}}}[\rho_c(t)]=\rho(t)$, so $\rho(t)=\mathbb{E}_{p_{\text{true}}}[\rho_c(t)]=\mathbb{E}_{p_{\text{ost}}}[\breve{\rho}_c(t)]$. In this sense, we can follow the evolution of the unconditional state from the non-normalized state evolution.

Likewise, in the case of photo-detection, the results in Eq. (II.34) hold and we can repeat the same kind of linearization method used in the previous paragraph. Therefore, one can obtain

$$\mathbb{E}_{p_{\text{ost}}}[\mathrm{d}N_t]=\mu' k\mathrm{d}t, \tag{II.35}$$

by replacing $\left\langle c^\dagger c\right\rangle_t$ by μ' in the photo-detection SME, i.e., in Eq. (II.12). The natural choice seems to be $\mu'=1$, so that the expectation value of $\mathrm{d}N_t$ (calculated with p_{ost}) is simply $k\mathrm{d}t$. It follows that the linear SME for photo-detection is

$$\mathrm{d}\breve{\rho}_c=-i[H_S,\breve{\rho}_c]\mathrm{d}t-\frac{k}{2}\Big\{c^\dagger c,\breve{\rho}_c\Big\}\mathrm{d}t+k\breve{\rho}_c\mathrm{d}t+\Big(c\breve{\rho}_c c^\dagger-\breve{\rho}_c\Big)\mathrm{d}N_t. \tag{II.36}$$

In the simulations we present in the last section of this chapter, we could use these linear quantum trajectories. However, we chose to use a Monte Carlo method that will be introduced in the following section, also, we will use a non-linear solver available in the `QuantumToolbox.jl` package.

II.6 Evolution in terms of Kraus operators

In the literature, one of the best solutions to simulate the photo-detection process, developed in Section II.3, is the so-called Monte Carlo method [38]. Nevertheless, we need to use Kraus operators in order to find the expression needed in the Monte Carlo simulation. In particular, the results of the QMT Section I.2 are going to be used hereafter.

II.6.1 Photo-detection

We define the Kraus operator yielding the result 0 as

$$K_0(\mathrm{d}t)=\mathbb{I}-\left(k\frac{c^\dagger c}{2}+iH_S\right)\mathrm{d}t, \tag{II.37}$$

[9]We can explicit the expression of the expected value of a continuous random variable, such that $X:\Omega\to\mathbb{R}$ with probability density function p, as $\mathbb{E}_p[X]=\int_\Omega X\mathrm{d}P=\int_{\mathbb{R}}xp(x)\mathrm{d}x$. In the case of the states (normalized or not), we should integrate over all possible trajectories (so every possible measurement outcome).

together with its adjoint $K_0^\dagger(\mathrm{d}t)$. For the result 1, we define

$$K_1(\mathrm{d}t) = \sqrt{k\mathrm{d}t}c, \tag{II.38}$$

and its adjoint $K_1^\dagger(\mathrm{d}t)$. These two Kraus operators should yield the definition of a POVM. We recall that a POVM must respect $\sum_n \Pi_n = \Pi_0 + \Pi_1 = \mathbb{I}$. In particular with the Kraus operators, we have that $\Pi_n = K_n^\dagger K_n$. However, using the definitions of the Kraus operators proposed in Eqs. (II.37,II.38) we obtain

$$K_0^\dagger(\mathrm{d}t)K_0(\mathrm{d}t) + K_1^\dagger(\mathrm{d}t)K_1(\mathrm{d}t) = \mathbb{I} + o(\mathrm{d}t). \tag{II.39}$$

Indeed, this last equation is verified at fist order in $\mathrm{d}t$ only, which is sufficient. Thus, these Kraus operators form a valid POVM. We can look at the probabilities to find each outcome in the Kraus formalism:

$$p_0(\mathrm{d}t) = \mathrm{Tr}\left[K_0^\dagger(\mathrm{d}t)\rho_c(t)K_0(\mathrm{d}t)\right] = 1 - k\left\langle c^\dagger c\right\rangle_t \mathrm{d}t + o(\mathrm{d}t), \tag{II.40}$$

$$p_1(\mathrm{d}t) = \mathrm{Tr}\left[K_1^\dagger(\mathrm{d}t)\rho_c(t)K_1(\mathrm{d}t)\right] = k\left\langle c^\dagger c\right\rangle_t \mathrm{d}t + o(\mathrm{d}t). \tag{II.41}$$

Therefore, we can trivially find that $p_0(\mathrm{d}t)+p_1(\mathrm{d}t) = 1+o(\mathrm{d}t)$. For the same reasons as aforementioned, the expected relation $p_0 + p_1 = 1$ is verified up to $\mathrm{d}t$, which is sufficient. Specifically, it is worthwhile to observe from Eq. (II.10) that $p_1(\mathrm{d}t) = \mathbb{E}[\mathrm{d}N_t] = \mathbb{P}[\mathrm{d}N_t = 1]$, which means that the event "applying Kraus operator K_1" is the same as the event "$\mathrm{d}N_t = 1$". As a consequence, we can rewrite the evolved conditional state in analogy with the approach used to obtain Eq. (II.11) but with Kraus operators this time, giving

$$\rho_c(t+\mathrm{d}t) = \underbrace{\frac{K_0(\mathrm{d}t)\rho_c(t)K_0^\dagger(\mathrm{d}t)}{p_0(\mathrm{d}t)}}_{\rho_0(t+\mathrm{d}t)}(1-\mathrm{d}N_t) + \underbrace{\frac{K_1(\mathrm{d}t)\rho_c(t)K_1^\dagger(\mathrm{d}t)}{p_1(\mathrm{d}t)}}_{\rho_1(t+\mathrm{d}t)}\mathrm{d}N_t. \tag{II.42}$$

This expression can be developed up to $\mathrm{d}t$, in the same way as in Section II.3, to find the SME, i.e. Eq. (II.12).

This new formulation is particularly interesting because the unnormalized expressions of evolved states (depending on each outcome) are

$$\tilde{\rho}_0(t+\mathrm{d}t) = K_0(\mathrm{d}t)\tilde{\rho}_c(t)K_0^\dagger(\mathrm{d}t), \qquad \tilde{\rho}_1(t+\mathrm{d}t) = K_1(\mathrm{d}t)\tilde{\rho}_c(t)K_1^\dagger(\mathrm{d}t). \tag{II.43}$$

Explicitly, by using Eqs (II.37,II.38), we get

$$\tilde{\rho}_0(t+\mathrm{d}t) = \tilde{\rho}_c(t) - \frac{k}{2}\left\{c^\dagger c, \tilde{\rho}_c(t)\right\}\mathrm{d}t - i[H_S, \tilde{\rho}_c(t)]\mathrm{d}t, \qquad \tilde{\rho}_1(t+\mathrm{d}t) = kc\tilde{\rho}_c(t)c^\dagger\mathrm{d}t. \tag{II.44}$$

The evolved system is in the state $\tilde{\rho}_0(t+\mathrm{d}t)$ with probability $p_0(\mathrm{d}t)$ or in the state $\tilde{\rho}_1(t+\mathrm{d}t)$ with probability $p_1(\mathrm{d}t)$. Moreover, we have here that $p_0(\mathrm{d}t) \gg p_1(\mathrm{d}t)$ in the infinitesimal limit and so the system should evolve almost every time according to the outcome 0 (at best it should be verified in average). When the state evolves along the 0 outcome only, we call this the "no-jump" evolution. In the following, we will be interested into simulating SSEs for simplicity and rapidity. Therefore, here are the conditional evolved states for each outcomes

$$\left|\tilde{\psi}_0(t+\mathrm{d}t)\right\rangle = K_0(\mathrm{d}t)\left|\tilde{\psi}_c(t)\right\rangle = \left[\mathbb{I} - \left(k\frac{c^\dagger c}{2} + iH_S\right)\mathrm{d}t\right]\left|\tilde{\psi}_c(t)\right\rangle, \tag{II.45}$$

$$\left|\tilde{\psi}_1(t+\mathrm{d}t)\right\rangle = K_1(\mathrm{d}t)\left|\tilde{\psi}_c(t)\right\rangle = \sqrt{k\mathrm{d}t}c\left|\tilde{\psi}_c(t)\right\rangle. \tag{II.46}$$

The Monte Carlo method [38] consists in allowing the state to evolve using the Kraus operator K_0 all the time, unless a jump occurs. Explicitly, the differential equation of no-jump evolution is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left|\tilde{\psi}_0(t)\right\rangle = -\left(k\frac{c^\dagger c}{2} + iH_S\right)\left|\tilde{\psi}_0(t)\right\rangle. \tag{II.47}$$

To determine when a jump occurs, we first need to pick a random number R before simulating the no-jump evolution. The effect of the non-normalized differential equation is notably to diminish the value of the normalization with respect to time. Indeed,

$$\mathrm{d}\left\langle\tilde{\psi}_0(t)\middle|\tilde{\psi}_0(t)\right\rangle = -k\left\langle\tilde{\psi}_0(t)\middle|c^\dagger c\middle|\tilde{\psi}_0(t)\right\rangle\mathrm{d}t. \tag{II.48}$$

The Monte Carlo process uses this equation to simulate the non-jump evolution up to a time T such that $\left\langle \tilde{\psi}_0(T) \middle| \tilde{\psi}_0(T) \right\rangle = R$. Indeed, the state's norm progressively diminishes from 1 (initially normalized state) to R following the no-jump evolution. The time T is the "jump" time at which we must apply the evolution using Eq. (II.46). The state is then normalized and we act again as in the beginning: picking a new random number R, the no-jump evolution, and so on. Note that applying the unnormalized evolution and then normalizing the state has the same effect as applying directly the normalized evolution. This last solution is the one that `QuantumToolbox.jl` employs in particular.

II.6.2 Homodyne detection

Following the same concepts than in the photo-detection case, we can define the Kraus operators responsible for the evolution according to the homodyne measuring process. In this sense, we define

$$K_{\mathrm{d}y_t} = \mathbb{I} - iH_S\mathrm{d}t - \frac{k}{2}c^\dagger c\mathrm{d}t + \sqrt{k}c\mathrm{d}y_t, \tag{II.49}$$

with its adjoint $K^\dagger_{\mathrm{d}y_t}$. Indeed, if we calculate the sum of POVM elements, we find

$$\int K_J^\dagger K_J \mathrm{d}J \neq \mathbb{I} + o(\mathrm{d}t), \tag{II.50}$$

where we denoted $J = \mathrm{d}y_t$ for convenience. This integral should be understood in the sense: "we sum the POVM elements over every possible outcome $\mathrm{d}y_t$ and over every possible trajectory". Indeed, the fact that $K_J^\dagger K_J = \mathbb{I} + \sqrt{k}(ce^{-i\theta} + c^\dagger e^{i\theta})J + o(dt)$ is the reason why the normalization of POVM fails here. Thus, to obtain the correct normalization, one must calculate the integral according to the ostensible probability (see Section II.5), giving

$$\int K_J^\dagger K_J p_{\mathrm{ost}}(J)\mathrm{d}J = \mathbb{I} + o(\mathrm{d}t). \tag{II.51}$$

This involves a more general definition of Kraus operators than the one initially introduced in the QMT section. Indeed, it is different in the sense that the normalization is not respected according to the true probability p_{true}. The unnormalized evolved state is

$$\tilde{\rho}_c(t+\mathrm{d}t) = K_{\mathrm{d}y_t}\rho_c(t)K^\dagger_{\mathrm{d}y_t} = \breve{\rho}_c(t+\mathrm{d}t), \tag{II.52}$$

where $\breve{\rho}_c(t+\mathrm{d}t)$ is the solution of the unnormalized linear SME. This last equation is related to the use of p_{ost} to obtain Eq. (II.51) because we have explained in the linear quantum trajectory section that $\mathbb{E}_{p_{\mathrm{ost}}}[\breve{\rho}_c(t)] = \rho(t)$. Indeed, the homodyne SME can be obtained again by averaging over every possible trajectories the conditional state evolution of Eq (II.52).

II.7 Simulations

In this section, we will focus on two different systems under continuous measurement of their environment. The first system is a qubit (two-levels quantum system), while the second system is a harmonic oscillator (infinite levels).

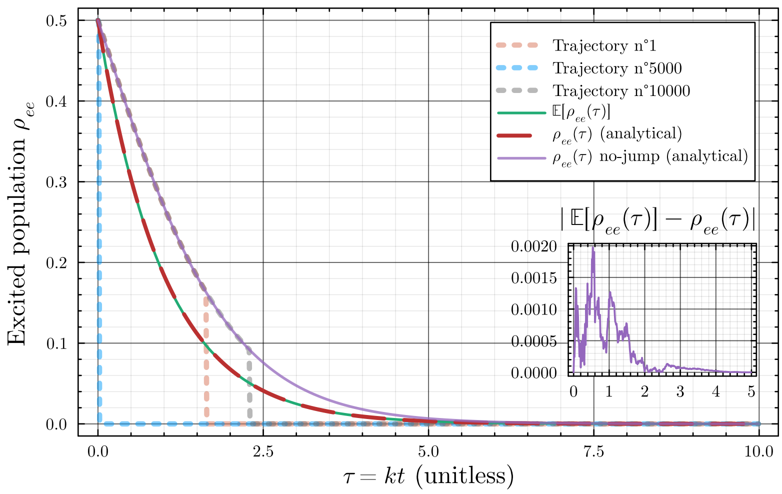
Before looking at the diverse simulations, we want to explain the conversion into dimensionless variables we will use for our simulations. By considering the Lindblad master equation, Eq. (I.39), we can introduce a dimensionless time $\tau = kt$ such that this equation rewrites as

$$\frac{\mathrm{d}\rho(t)}{\mathrm{d}\tau} = -\frac{i}{k}[H_S, \rho(t)] + \mathfrak{D}[c]\rho(t). \tag{II.53}$$

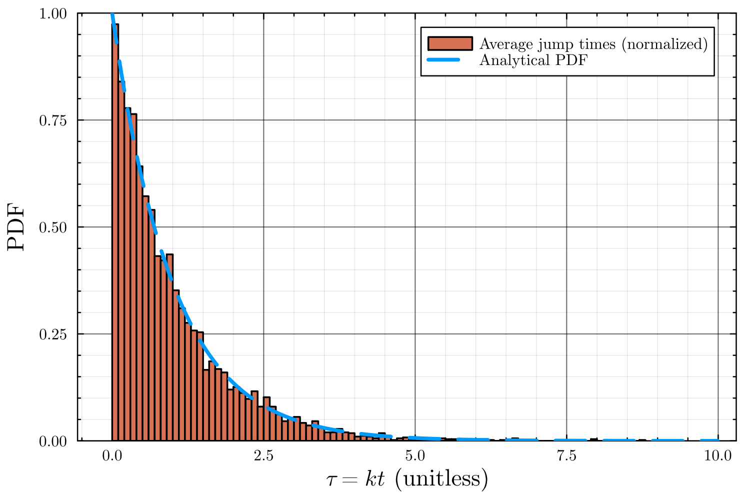
In any of the following cases (simulating a qubit or a harmonic oscillator), the Hamiltonian H_S is proportional to the natural frequency of the system denoted as ω_0. Hence, the first term of the Lindblad master equation yields the dimensionless frequency $\frac{\omega_0}{k}$. To respect the weak-coupling condition, it must verify $\omega_0 \gg k$, so, in the simulation we chose to set it to $\frac{\omega_0}{k} = 10$. We know from OQS theory that after some oscillations, the system reaches the stationary state. Thus, our simulations will span over $\tau \in [0, 10]$ to be close enough to the stationary state at the end. This reasoning stands for SME simulations because we know that the average over every possible trajectories gives back the Lindblad unconditional evolution. We precise that the environments of the following systems are always at the zero-temperature limit, i.e., we consider the vacuum state of the environmental oscillator at each time $|0\rangle\langle 0|_t$.

II.7.1 The qubit under photo-detection of its environment

The simulation of a qubit under photo-detection of its environment is realized by implementing the Monte Carlo simulation method into `Julia`. The package `QuantumToolbox.jl` has a dedicated function called `mcsolve` which does it fine. This function requires in particular to specify the Hamiltonian of the system S, the initial state (as a vector or a density operator), the number of trajectories to simulate (in order to average over all trajectories) and the operators to apply.



(a) The evolution of the excited population of the qubit with respect to dimensionless time τ compared to the unconditional evolution of the excited population.



(b) The histogram of simulation jump times T of every simulations compared to the analytical Lindblad PDF.

Figure 4: Results obtained from the Monte Carlo simulation of a qubit under continuous photo-detection of its environment.

To work with the qubit, we briefly detail the notations and operators used in the description of two-level systems. Firstly, the eigenstates of the Hilbert space $\mathcal{H}_S$ are denoted as $|e\rangle = \begin{pmatrix}1\\0\end{pmatrix}$ and $|g\rangle = \begin{pmatrix}0\\1\end{pmatrix}$, where e is the excited state and g is the ground state of the qubit. Secondly, the three Pauli matrices that form (together with the identity $\mathbb{I}$) an eigenbasis of $\mathcal{H}_S$ are denoted as

$$\sigma_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}. \tag{II.54}$$

We will also use the ladder operators

$$\sigma_- = \begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}, \quad \sigma_+ = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}, \tag{II.55}$$

which act respectively as a lowering and a raising operator, inducing transitions from the excited to the ground state and vice-versa. Notice that in a two-levels system, the density matrix is simply

$$\rho = \begin{pmatrix}\rho_{ee} & \rho_{eg}\\ \rho_{ge} & \rho_{gg}\end{pmatrix}, \tag{II.56}$$

where ρ_{ee} and ρ_{gg} are respectively the excited and ground populations, and where ρ_{eg} and ρ_{ge} are the correlations. Notice that the definition of a density matrix, see Section I.1.3, implies that $\rho_{ee} + \rho_{gg} = 1$, that $\rho_{ge} = \rho_{eg}^*$, and that $\rho_{ee}\rho_{gg} \geq |\rho_{eg}|^2$ in particular.

In the case of the qubit, the no-jump evolution Eq (II.47) becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\left|\tilde{\psi}_0(t)\right\rangle = -\left(\frac{k}{2}\sigma_+\sigma_- + i\frac{\omega_0}{2}\sigma_z\right)\left|\tilde{\psi}_0(t)\right\rangle, \tag{II.57}$$

where $H_S = \frac{\omega_0}{2}\sigma_z$. In anticipation of the simulation, we have seen theoretically in Section II.3 that the key quantity available to an experimentalist is the set of photo-detection outcomes $\{\mathrm{d}N_t\}_t$. The stochastic expectation value of this random variable has been shown to be $\mathbb{E}[\mathrm{d}N_t] = k\left\langle c^\dagger c\right\rangle_t \mathrm{d}t = \rho_{ee}^{(c)}(\tau)\mathrm{d}\tau$, where we denoted the conditional property of this state as (c) for convenience. This average is only stochastic on a single trajectory and we can average over every trajectories to find that $\mathbb{E}_{\text{trajs}}[\mathrm{d}N_\tau] = \rho_{ee}(\tau)\mathrm{d}\tau$, where the state is now unconditional. This property is used in the Figure 4b to build the so-called analytical Probability Density Function (PDF) from the Lindblad evolution. Indeed, very generally from the OQS theory, one can show that $\rho_{ee}(\tau) = \rho_{ee}(0)e^{-\tau}$ [3]. Therefore by normalizing over the simulation period so that the area under the curve is exactly 1, we find that

$$\mathrm{PDF}(\tau) = \frac{e^{-\tau}}{1 - e^{-\tau_{\max}}}. \tag{II.58}$$

This PDF is actually independent of the system choice and will be the same for the harmonic oscillator. This is straightforward to show when replacing the generic c operator by the annihilation operator a, and by knowing that $\langle \hat{n}(\tau)\rangle = \left\langle a^\dagger a\right\rangle(\tau) = \langle n(0)\rangle\, e^{-\tau}$ [3].

The average excited population of every simulated trajectories is compared to the unconditional evolution of the excited population in Figure 4a. We have simulated 10,000 trajectories to obtain this graph within the `MCQubit.jl` program (all the programs are available on the GitHub repository LIEN). In this program, we chose to set the initial state as $|\psi(\tau = 0)\rangle = \frac{|e\rangle + |g\rangle}{\sqrt{2}}$, and we can observe that the excited population is indeed 0.5 at the beginning on Figure 4a. As expected, the average of the 10,000 trajectories follows the unconditional trajectory. The error inset plot in Figure 4a is showing that the difference between these two curves is small enough to consider the result valid. We expect the difference to decrease as the number of trajectory increases. Ideally, we should find exactly the same curves as the number of trajectory tends to infinity.

Moreover, some single trajectories are plotted on this figure to show their behavior. Theoretically, it is possible that a trajectory never jumps because of the superposition of states initially set. Indeed, if the random number R picked is very small, the system might have time to purify to the ground state before the norm diminishes enough. Specifically, we know that SMes conserve the purity of states and we can observe that the final excited population is 0, meaning that the final state is $|g\rangle$. Hence, the norm is 1 at the end and obviously cannot be smaller than R. This reasoning seems to imply a competition between the environmental decoherence governed by the Lindblad master equation and the state purification induced by the continuous monitoring.

Furthermore, the curve labeled "$\rho_{ee}(\tau)$ no-jump (analytical)" is the evolution of the state without never jumping. To reach the analytical expression of this evolution, we simply have to solve Eq. (II.57). For a generic superposition $|\psi(\tau)\rangle = \alpha|e\rangle + \beta|g\rangle$, we find

$$\left|\tilde{\psi}_0(t)\right\rangle = \alpha e^{-i\frac{\omega_0}{2}(t-t_0)} e^{-\frac{k}{2}(t-t_0)}|e\rangle + \beta e^{i\frac{\omega_0}{2}(t-t_0)}|g\rangle . \tag{II.59}$$

We can then calculate the square of the state's norm in order to obtain the expression of the mean excited population within the no-jump evolution. We easily find that

$$\left\langle\tilde{\psi}_0(t)\middle|\tilde{\psi}_0(t)\right\rangle = |\alpha|^2 e^{-k(t-t_0)} + |\beta|^2 \quad \Longrightarrow \quad |\langle e|\psi_0(t)\rangle|^2 = \frac{1}{1+(|\beta|^2/|\alpha|^2)e^{k(t-t_0)}}. \tag{II.60}$$

Therefore in our case ($\alpha = \beta = \sqrt{2}$), we have that the no-jump evolution is given by

$$\rho_{ee}^{\text{no-jump}}(t) = \frac{1}{1+e^{kt}}, \tag{II.61}$$

with the initial condition $\rho_{ee}^{\text{no-jump}}(0) = \frac{1}{2}$ as expected. This last equation explains the origin of the purple curve in Figure 4a. We can observe that the no-jump evolution does not coincide with the unconditional evolution. Indeed, the presence of the photo-detector has an impact on the dynamics even if any photon is detected.

II.7.2 The harmonic oscillator under photo-detection of its environment

This simulation is embedded within the `MCHarm.jl` program with the same function `mcsolve`, as in the last subsection. The initial state is $|\psi(0)\rangle = |\alpha_0\rangle$, where α_0=2. This state is a coherent state [43], by using the Fock states $\{n\}_{n\in\mathbb{N}}$, they are defined as

$$|\alpha_0\rangle = e^{-\frac{|\alpha_0|^2}{2}} \sum_{n=0}^{+\infty} \frac{\alpha_0^n}{\sqrt{n!}} |n\rangle . \tag{II.62}$$

A key property of this kind of state is in particular that $a|\alpha_0\rangle = \alpha_0|\alpha_0\rangle$, where a is the annihilation operator of the system. Another property is that $\langle\alpha_0|\hat{n}|\alpha_0\rangle = |\alpha_0|^2$, which is thus the initial population of the system. The Hamiltonian of the system is now $H_S = \omega_0 a^\dagger a$, and the no-jump evolution followed by the Monte Carlo process is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left|\tilde{\psi}_0(t)\right\rangle = -\left(\frac{k}{2}a^\dagger a + i\omega_0 a^\dagger a\right)\left|\tilde{\psi}_0(t)\right\rangle . \tag{II.63}$$

As explained in the last subsection, the PDF in Figure 5b is the same as in the case of a qubit, see Eq. (II.58). Therefore, this figure is very similar to the Figure 4a as expected. Indeed, the average over all trajectories of the jump times must give the unconditional PDF (which is thus the same for both systems).

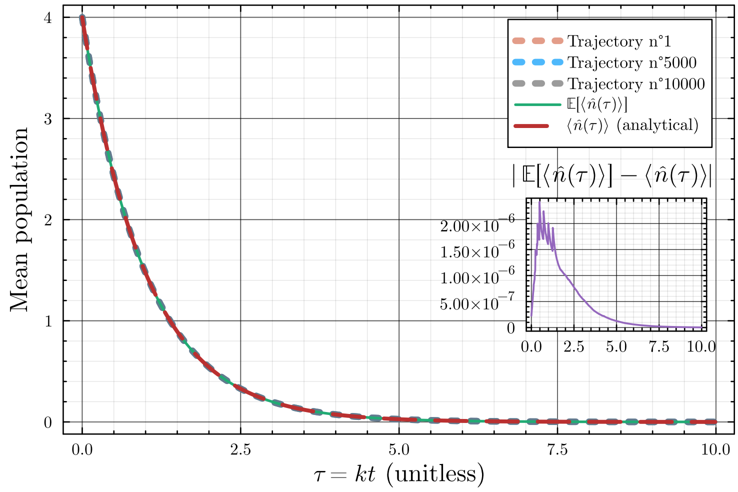
The unconditional mean population is given by

$$\langle\hat{n}(\tau)\rangle = \langle\hat{n}(0)\rangle e^{-\tau} = |\alpha_0|^2 e^{-\tau}. \tag{II.64}$$

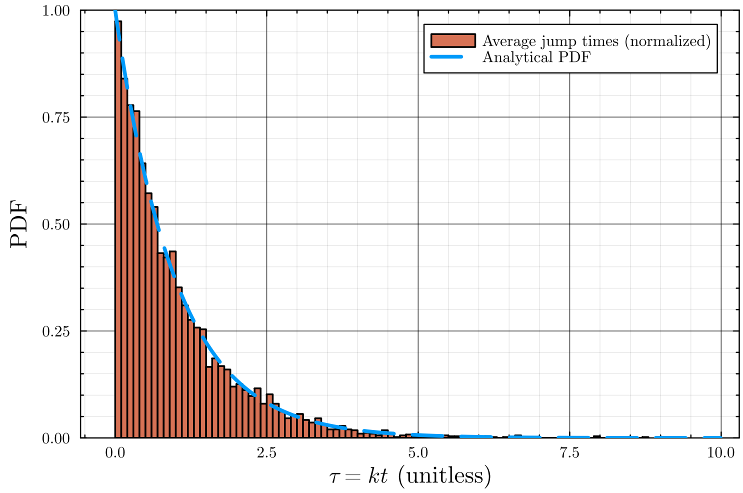
This evolution is represented by the dash-curve labeled by "$\langle\hat{n}(\tau)\rangle$ (analytical)" in Figure 5a. Moreover, we represent some single trajectories with dotted lines, which coincide with the unconditional dynamics in the case of a coherent initial state. To obtain the simulation curve, we averaged 10,000 trajectories and we see that, as expected, the average evolution curve superposes with the unconditional evolution. However, compared to the qubit case, we see that the errors (present in the inset of Figure 5a) are way more small (around 10^{-6} order of magnitude compared to 10^{-3} at worst). To have a hint on why the error is so low in this case, we might want to calculate the no-jump mean population evolution.

By solving Eq. (II.63), we find

$$\left|\tilde{\psi}_0(t)\right\rangle = e^{-i\omega_0 a^\dagger a(t-t_0)} e^{-\frac{k a^\dagger a}{2}(t-t_0)} |\alpha_0\rangle , \tag{II.65}$$



(a) The evolution of the mean population of the harmonic oscillator with respect to dimensionless time τ compared to the unconditional evolution of the mean population.



(b) The histogram of simulation jump times T of every simulations compared to the analytical Lindblad PDF.

Figure 5: Results obtained from the Monte Carlo simulation of a harmonic oscillator under continuous photodetection of its environment.

which will be used to calculate $\langle \psi_0(\tau)|\hat{n}|\psi_0(\tau)\rangle$ after some steps. To do so, we might want to express this evolved state as a coherent state before any calculation. Indeed,

$$\left|\tilde{\psi}_0(t)\right\rangle = e^{\gamma(t)a^\dagger a} |\alpha_0\rangle\,, \tag{II.66}$$

where we used $\gamma(t) = -(i\omega_0 + \frac{k}{2})(t - t_0)$ as a shorthand notation. By virtue of Eq. (II.62), this state can be expressed as

$$\left|\tilde{\psi}_0(t)\right\rangle = e^{-\frac{|\alpha_0|^2}{2}} \sum_{n=0}^{+\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{\gamma(t)a^\dagger a} |n\rangle = e^{-\frac{|\alpha_0|^2}{2}} \sum_{n=0}^{+\infty} \frac{\left(\alpha_0 e^{\gamma(t)}\right)^n}{\sqrt{n!}} |n\rangle\,. \tag{II.67}$$

To force the apparition of a coherent state in the last expression, we set

$$\alpha(t) = \alpha_0 e^{\gamma(t)} = \alpha_0 e^{-i\omega_0(t-t_0)} e^{-\frac{k}{2}(t-t_0)}. \tag{II.68}$$

Using the calculation trick of multiplying a term by its inverse, we eventually find

$$\left|\tilde{\psi}_0(t)\right\rangle = e^{\frac{|\alpha(t)|^2}{2}} e^{-\frac{|\alpha(t)|^2}{2}} \left|\tilde{\psi}_0(t)\right\rangle = e^{-\frac{|\alpha_0|-|\alpha(t)|^2}{2}} |\alpha(t)\rangle\,. \tag{II.69}$$

This state is easy to normalize since $\langle\beta|\beta\rangle = 1$ for any coherent state $|\beta\rangle$. Therefore, we proved that starting with a coherent state $|\alpha_0\rangle$ assures that

$$|\psi_0(t)\rangle = |\alpha(t)\rangle = \left|\alpha_0 e^{-i\omega_0(t-t_0)} e^{-\frac{k}{2}(t-t_0)}\right\rangle. \tag{II.70}$$

The mean population can be simply evaluated thanks to $\langle\beta|\hat{n}|\beta\rangle = |\beta|^2$ for any coherent state $|\beta\rangle$. This relation implies that

$$\langle\psi_0(t)|\hat{n}|\psi_0(t)\rangle = |\alpha_0|^2 e^{-k(t-t_0)}. \tag{II.71}$$

If we now compare this result with the unconditional evolution in Eq. (II.64), we realize that they are exactly the same. This result explains why the errors between the average curve, the unconditional curve and the no-jump curve is so small. We seem to have found an initial state which is not bothered by the photo-detection measurement of its environment.

II.7.3 The qubit under homodyne detection of its environment

For the homodyne measuring process we simulate the evolution of the conditional state with the function `smesolve` from `QuantumToolbox.jl`. The Hamiltonians and initial states are the same for each system but we chose to work with density matrices in the programs (here `XHomodyneQubit.jl` and `PHomodyneQubit.jl`) for demonstrative purpose.

In the qubit case, one quickly realizes that the 0 and $\pi/2$ quadratures are interesting operators to measure since they are linked to the correlation of the state. To show this, we need to work on the expression of $\langle\sigma_\theta\rangle = \mathrm{Tr}[\rho(t)\sigma_\theta]$ with a general density matrix ρ, in which we define the qubit's quadrature operator by

$$\sigma_\theta = \frac{\sigma_- e^{-i\theta} + \sigma_+ e^{i\theta}}{\sqrt{2}}. \tag{II.72}$$

This expression can be developed to let the coherences appear explicitly, yielding

$$\langle\sigma_\theta\rangle = \frac{\rho_{eg} e^{-i\theta} + \rho_{ge} e^{i\theta}}{\sqrt{2}} = |\rho_{eg}| \frac{e^{i(\varphi-\theta)} + e^{i(\theta-\varphi)}}{\sqrt{2}}, \tag{II.73}$$

where we used the exponential notation for the coherences (which are complex scalars). In particular, since $\rho_{ge} = \rho_{eg}^*$ we can write that $\rho_{eg} = |\rho_{eg}|e^{i\varphi}$ and $\rho_{ge} = |\rho_{eg}|e^{-i\varphi}$. We can recognize the expression of a cosinus, explicitly

$$\langle\sigma_\theta\rangle = \sqrt{2}|\rho_{eg}|\cos(\varphi - \theta). \tag{II.74}$$

To summarize, the aforementioned quadratures are therefore

$$\langle\sigma_0\rangle = \sqrt{2}|\rho_{eg}|\cos(\varphi) = \sqrt{2}\mathrm{Re}\{\rho_{eg}\}, \quad \langle\sigma_{\pi/2}\rangle = \sqrt{2}|\rho_{eg}|\sin(\varphi) = \sqrt{2}\mathrm{Im}\{\rho_{eg}\}. \tag{II.75}$$

The simulations yield the graphs in Figure 6 where we can compare the averaged quadratures with the analytical unconditional quadratures. To find the expressions of these analytical curves (one per quadrature), we used the OQS theory [3], which gives

$$\rho_{eg}(t) = \rho_{eg}(t_0) e^{-\left(\frac{k}{2} + i\omega_0\right)(t-t_0)}. \tag{II.76}$$

Expanding the complex exponential with the Euler formula gives

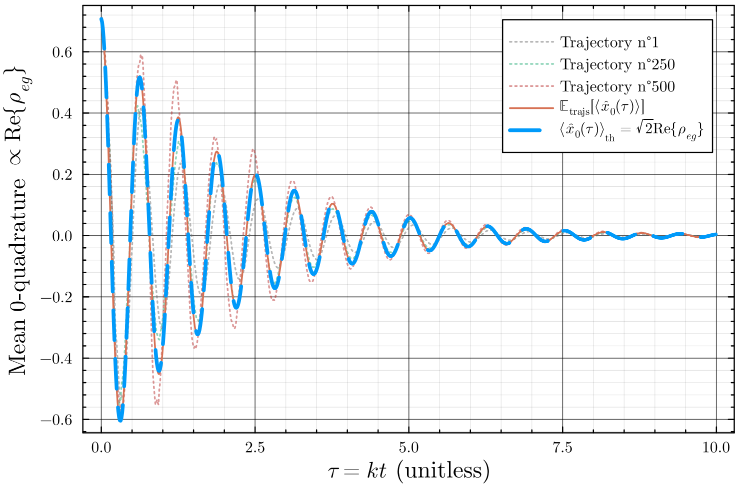
$$\rho_{eg}(t) = \rho_{eg}(t_0) e^{-\frac{k(t-t_0)}{2}} \{\cos[\omega_0(t-t_0)] - i\sin[\omega_0(t-t_0)]\}. \tag{II.77}$$

The initial state being the density matrix built from $|\psi(0)\rangle = \frac{|e\rangle + |g\rangle}{\sqrt{2}}$, it gives

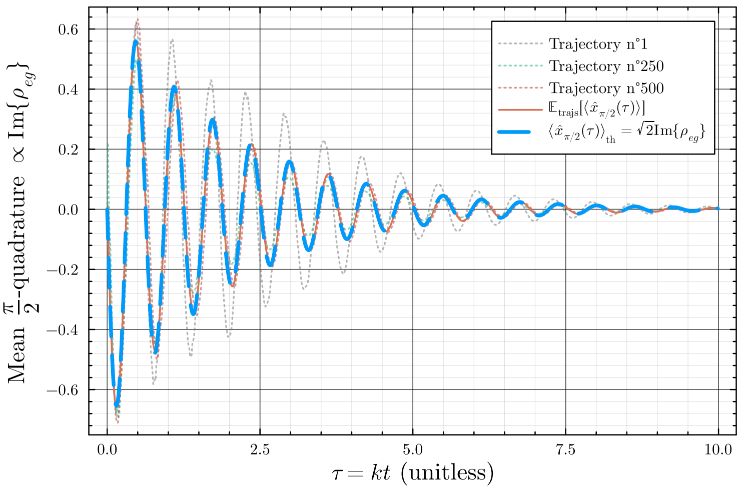
$$\begin{cases} \sqrt{2}\mathrm{Re}\{\rho_{eg}(\tau)\} = \frac{e^{-\frac{k\tau}{2}}}{\sqrt{2}}\cos(\Omega\tau) \\ \sqrt{2}\mathrm{Im}\{\rho_{eg}(\tau)\} = -\frac{e^{-\frac{k\tau}{2}}}{\sqrt{2}}\sin(\Omega\tau) \end{cases}. \tag{II.78}$$

This system of equation give the dashed blue curves on graphs of Figure 6, while the averaged trajectories are the two solid orange curves.

For consistency, we plot the absolute difference between a single trajectory and the unconditional evolution (solid grey curve), and the absolute difference between the averaged trajectories and the unconditional trajectory (solid orange curve) present on Figure 7 for each quadrature. These graphs emphasize the fact that the averaged trajectory coincides with the unconditional Lindblad master equation trajectory. It will compared with the following results in the case of the harmonic oscillator instead of the qubit.

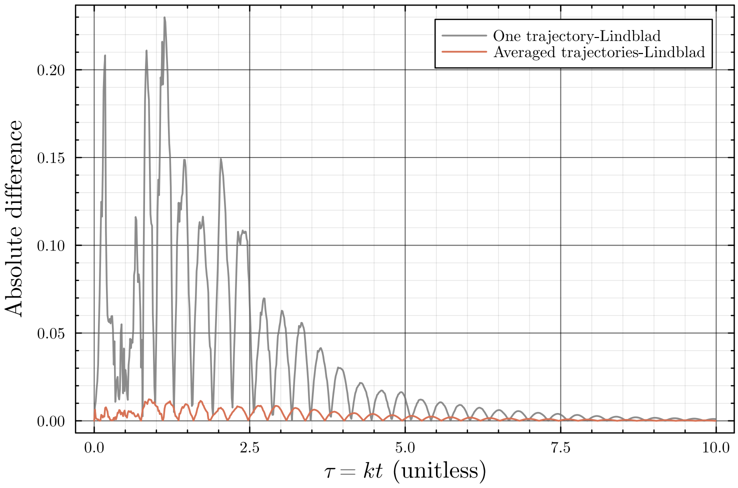


(a) The evolution of $\sqrt{2}\text{Re}\{\rho_{eg}\}$ of the qubit with respect to dimensionless time τ compared to the unconditional evolution of the same quantity.

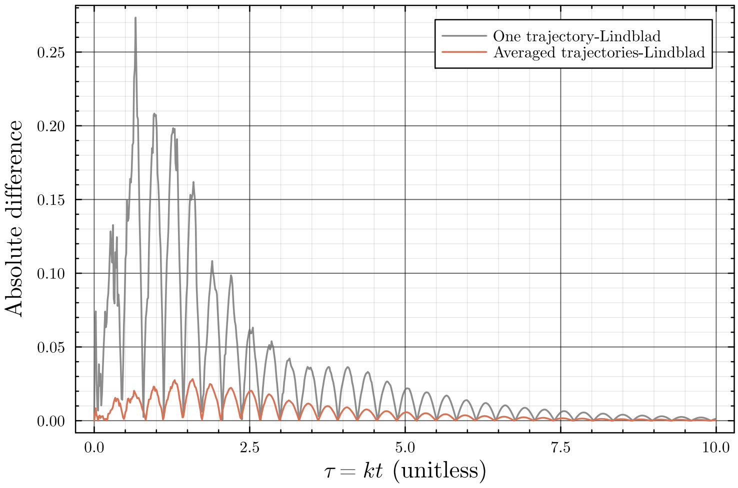


(b) The evolution of $\sqrt{2}\text{Im}\{\rho_{eg}\}$ of the qubit with respect to dimensionless time τ compared to the unconditional evolution of the same quantity.

Figure 6: The evolution of two qubit quadratures obtained by simulating continuous homodyne detection of the qubit's environment.



(a) Absolute difference between a single trajectory or the averaged trajectories with the unconditional trajectory for the 0 quadrature (proportional to the real part of ρ_{eg}).



(b) Absolute difference between a single trajectory or the averaged trajectories with the unconditional trajectory for the $\pi/2$ quadrature (proportional to the imaginary part of ρ_{eg}).

Figure 7: Graphs highlighting the convergence of averaged trajectories towards the unconditional trajectory, compared to a single trajectory, for the qubit under homodyne measurement of its environment.

II.7.4 The harmonic oscillator under homodyne detection of its environment

The case of the harmonic oscillator is similar to the last on. We only change the Hamiltonian and the initial system to be a coherent density matrix, in analogy of the settings in Section II.7.2 but with the density matrix formalism (for demonstrative purpose only). These steps are encoded in the program named `HomodyneHarmonic.jl`. Unlike the previous cases, we don't plot some single trajectories because they are extremely close to the unconditional/averaged trajectories, this is highlighted in the error plots of Figure 9. Indeed, compared to the Figure 7, we observe a difference in the order of magnitude of errors (the qubit single-trajectory errors are around 10^{-1} to 10^{-2}, while for the harmonic oscillator they are around 10^{-4}).

To obtain the analytical expression for the unconditional trajectory, we need an important result of OQS theory: a coherent state stays coherent under the Lindblad master equation evolution [3]. The expression of the state at time t is

$$\alpha(t) = e^{-\left(i\omega_0 + \frac{k}{2}\right)(t-t_0)}\alpha(t_0). \tag{II.79}$$

By defining the quadrature operator of the harmonic oscillator as

$$a_\theta = \frac{ae^{-i\theta} + a^\dagger e^{i\theta}}{\sqrt{2}}, \tag{II.80}$$

we can analytically calculate the mean value of this operator. Explicitly,

$$\langle\alpha(t)|a_\theta|\alpha(t)\rangle = \frac{\langle\alpha(t)|a|\alpha(t)\rangle\, e^{-i\theta} + \langle\alpha(t)|a^\dagger|\alpha(t)\rangle\, e^{i\theta}}{\sqrt{2}}. \tag{II.81}$$

The fundamental relation $a\,|\alpha\rangle = \alpha\,|\alpha\rangle$ allows to simplify the previous equation into

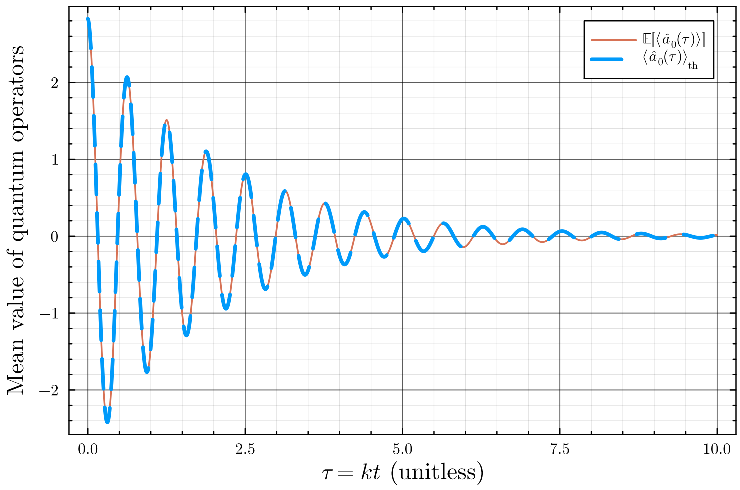
$$\langle a_\theta(t)\rangle = \frac{\alpha(t)e^{-i\theta} + \alpha(t)^* e^{i\theta}}{\sqrt{2}} = |\alpha(t_0)|\frac{e^{-i[\omega_0(t-t0)+\theta]} + e^{i[\omega_0(t-t0)+\theta]}}{\sqrt{2}} e^{-\frac{k(t-t_0)}{2}}. \tag{II.82}$$

By recognizing the expression of a cosinus, we obtain

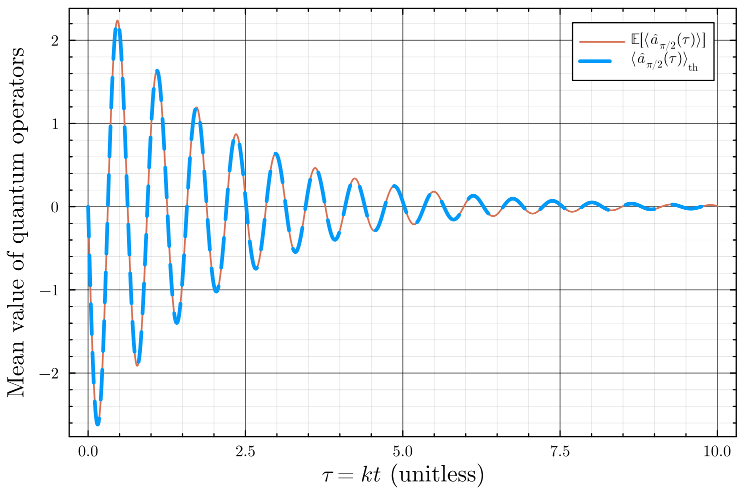
$$\langle a_\theta(t)\rangle = \sqrt{2}|\alpha_0|e^{-\frac{kt}{2}}\cos(\omega_0 t + \theta). \tag{II.83}$$

Since $\alpha_0 = 2$, the dashed blue curves present respectively in Figure 8a and Figure 8b are simply

$$\begin{cases} \langle a_0(t)\rangle = 2\sqrt{2}e^{-\frac{kt}{2}}\cos(\omega_0 t) \\ \left\langle a_{\pi/2}(t)\right\rangle = -2\sqrt{2}e^{-\frac{kt}{2}}\sin(\omega_0 t) \end{cases}. \tag{II.84}$$

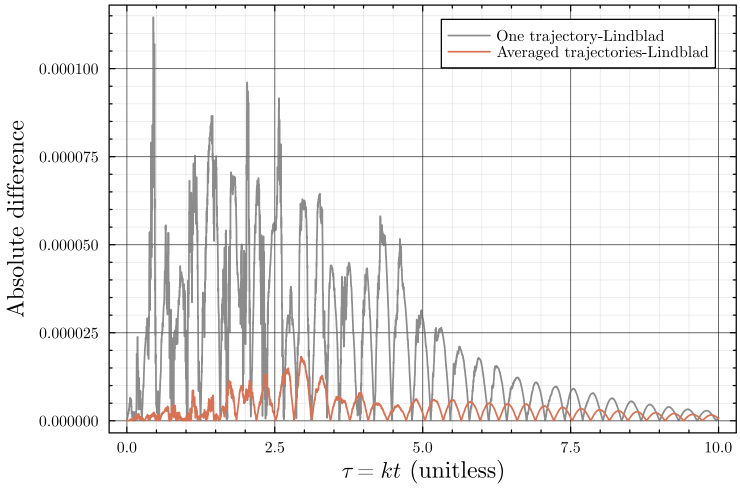


(a) The evolution of $\sqrt{2}\text{Re}\{\rho_{eg}\}$ of the qubit with respect to dimensionless time τ compared to the unconditional evolution of the same quantity.

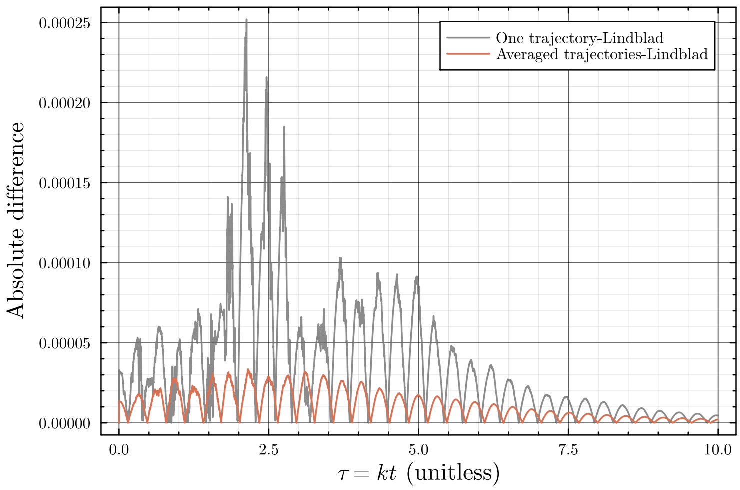


(b) The evolution of $\sqrt{2}\text{Im}\{\rho_{eg}\}$ of the qubit with respect to dimensionless time τ compared to the unconditional evolution of the same quantity.

Figure 8: The evolution of two qubit quadratures obtained by simulating continuous homodyne detection of the qubit's environment.



(a) Absolute difference between a single trajectory or the averaged trajectories with the unconditional trajectory for the 0 quadrature.



(b) Absolute difference between a single trajectory or the averaged trajectories with the unconditional trajectory for the $\pi/2$ quadrature.

Figure 9: Graphs highlighting the convergence of averaged trajectories towards the unconditional trajectory, compared to a single trajectory, for the harmonic oscillator under homodyne measurement of its environment.

Conclusion

This work, carried out in the framework of an internship during the final months of a Master's degree, has been dedicated to understanding and studying the evolution of quantum open systems under continuous measurement of their environment. We therefore described the fundamental theories of closed quantum systems up to the unconditional master equation describing the evolution of open quantum systems. In particular, the derivation of the Lindblad master equation presented in the first chapter was performed within the input-output formalism under the Born-Markov approximations.

The heart of our study aimed on the impact of continuous monitoring of the environment of open quantum systems in the second chapter. This impact is specifically crystallized in the conditional trajectories depending on which process of measurement is made on the environment. We studied two of these processes in details, the first being described by a jump-like output signal (photo-detection), while the second one being characterized by a continuous diffusive signal (homodyne detection). The studied processes have been modeled with stochastic processes, namely a Poisson process and a Wiener process. We also extended this study to the theoretical expressions allowing to simulate trajectories, whose underlying stochastic differential equations (called stochastic master equations) are initially non-linear and challenging to solve. This has been made possible by deriving linear quantum trajectories and by using the Kraus operators formalism. Finally, we simulated some trajectories of a qubit and a harmonic oscillator under the continuous monitoring of their environment. These simulations clearly highlighted the validity of the theoretical results since the averaged trajectory is perfectly coincides with the unconditional trajectory (when the number of trajectories tends to infinity). This property explains precisely why the derivation of a specific stochastic master equation is called an "unravelling" of the Lindblad unconditional master equation.

Beyond the theoretical results established in this report, several directions can extend this work. While our study focused exclusively on a vacuum bath, a natural next step would be to incorporate thermal environments [44]. Furthermore, introducing more realistic descriptions would bridge the gap toward experimental reality, such as accounting for imperfect measurement processes and detector inefficiencies [5]. In the same vein, the framework of homodyne detection can be generalized to the so-called "heterodyne" or even "general-dyne" detection schemes [5, 40, 45]. Beyond the standard systems considered here, the scientific community studied more complex system, e.g., a single nonlinear Kerr resonator whose Hamiltonian features quadratic driving terms [46]. Finally, the theory of continuous quantum measurement could be naturally integrated into the (huge) framework of quantum estimation and metrology [33, 34].

Appendix A

Representations of quantum mechanics

In quantum mechanics we define the expectation value of an operator O by

$$\langle O \rangle_\psi = \langle \psi | O | \psi \rangle ,$$
$$\langle O \rangle_\rho = \mathrm{Tr}\left[\rho O\right].$$

A.1 Schrödinger and Heisenberg pictures

In the Schrödinger picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$, where $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. Nevertheless, the following results can be derived in the general case of a time dependent Hamiltonian, [3]. Considering the average value of an observable $A_S(t)$ in the Schrödinger picture (in this picture, the time-dependence of an observable can only be explicit), we find

$$\begin{aligned} \langle A(t) \rangle &= \langle \psi(t) | A_S(t) | \psi(t) \rangle \\ &= \langle \psi(t_0) | U^\dagger(t, t_0) A_S(t) U(t, t_0) | \psi(t_0) \rangle && \text{(A. 1)} \\ &= \langle A_H(t) \rangle . && \text{(A. 2)} \end{aligned}$$

This calculation naturally reveals the Heisenberg representation of observables and states. Indeed, we have that $|\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time t_0. Furthermore, we can link both representation at all time by calculating the evolution of an observable in the Heisenberg picture:

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}}. \tag{A. 3}$$

This equation leads to the famous *Ehrenfest theorem* analogous in quantum mechanics:

$$\boxed{\frac{\mathrm{d}\langle A \rangle}{\mathrm{d}t} = \frac{i}{\hbar}\langle [H(t), A(t)] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle}. \tag{A. 4}$$

We observe that if the observable does not depend on time explicitly (the second term is 0 in the last equations), then the dynamics of the operator is fully described in the Heisenberg picture.

A.2 Interaction picture

As mentioned in the Section I.3, we can represent the coupling between S and E with an interaction Hamiltonian such that

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{A. 1}$$

From now on, we denote the state of the total system as $\boldsymbol{\varrho}$. Knowing that $|\psi(t)\rangle = U(t, t_0)\,|\psi\rangle$, we can write that

$$\boldsymbol{\varrho}(t) = U(t, t_0)\boldsymbol{\varrho}(t_0)U^\dagger(t, t_0). \tag{A. 2}$$

The evolution operator U is the solution of the Schrödinger equation for the total system T. We know from Section I.1.1 that we can write the solution of the Schrödinger equation as an exponential when dealing with a time-independent Hamiltonian. However here, the total Hamiltonian H_T is obviously time-dependent (only H_0 is assumed constant in time). Nevertheless, a unitary solution still exists [3] and is denoted by U hereafter. the free system, governed by H_0, follows an unitary dynamics described by $U_0(t,t_0) = e^{-\frac{i}{\hbar}H_0(t-t_0)}$. Thus, we can define the interaction unitary operator U_I as

$$U_I(t,t_0) = U_0^\dagger(t,t_0)U(t,t_0) \tag{A. 3}$$

$$\Leftrightarrow U(t,t_0) = U_0(t,t_0)U_I(t,t_0) \tag{A. 4}$$

If we now try to calculate the average of an operator (in the total space $\mathcal{H}_T$) using the Schrödinger picture, we can find a new interesting relation by injecting the Eq. (A. 4) in the Eq. (A. 2) and then into the following average:

$$\begin{aligned}\langle O(t)\rangle &= \mathrm{Tr}\left[\boldsymbol{\varrho}(t)O(t)\right] \\ &= \mathrm{Tr}\,[\underbrace{U_I(t,t_0)\boldsymbol{\varrho}(t_0)U_I^\dagger(t,t_0)}_{\tilde{\boldsymbol{\varrho}}(t)}\,\underbrace{U_0^\dagger(t,t_0)O(t)U_0(t,t_0)}_{\tilde{O}(t)}].\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\boldsymbol{\varrho}}(t) = U_I(t,t_0)\boldsymbol{\varrho}(t_0)U_I^\dagger(t,t_0), \tag{A. 5}$$

$$\tilde{O}(t) = U_0^\dagger(t,t_0)O(t)U_0(t,t_0). \tag{A. 6}$$

We can now reverse the Eq. (A. 2) and then use the Eq. (A. 4) to get

$$\boldsymbol{\varrho}(t_0) = U^\dagger(t,t_0)\boldsymbol{\varrho}(t)U(t,t_0) = U_I^\dagger(t,t_0)U_0^\dagger(t,t_0)\boldsymbol{\varrho}(t)U_0(t,t_0)U_I(t,t_0),$$

and, by injecting this into the Eq. (A. 5), we get

$$\tilde{\boldsymbol{\varrho}}(t) = U_0^\dagger(t,t_0)\boldsymbol{\varrho}(t)U_0(t,t_0). \tag{A. 7}$$

This result is obvious because a density matrix is also an operator and thus respects the Eq. (A. 6). In the end, we find the results used in the main part of this document:

$$\boxed{\tilde{\boldsymbol{\varrho}}(t) = U_I(t,t_0)\boldsymbol{\varrho}(t_0)U_I^\dagger(t,t_0) = U_0^\dagger(t,t_0)\boldsymbol{\varrho}(t)U_0(t,t_0)}. \tag{A. 8}$$

By working on the Liouville-von Neumann equation, one can show that by jumping into the interaction picture the dynamics depends on another Hamiltonian:

$$\left(\frac{\mathrm{d}\boldsymbol{\varrho}}{\mathrm{d}t} = -\frac{i}{\hbar}[H_T,\boldsymbol{\varrho}]\right) \Leftrightarrow \left(\frac{\mathrm{d}\tilde{\boldsymbol{\varrho}}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I,\tilde{\boldsymbol{\varrho}}\right]\right), \tag{A. 9}$$

where $\tilde{H}_I = U_0^\dagger(t,t_0)H_IU_0(t,t_0)$. It is thus usual to work in this representation to get rid of free dynamics terms, and then jump back to the Schrödinger picture to recover the proper dynamics of the system.

Further discussion

We used in Eq. (I.33) the following approximation: $U_I(t+\mathrm{d}t,t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t}$, and we would like to explain it here. As seen in Eq. (A. 5), a state evolves in the interaction picture with a relation depending on U_I, which has been defined as $U_I = U_0^\dagger U$. It is possible to show from this definition that U_I is subject to the following differential equation

$$\frac{\mathrm{d}U_I}{\mathrm{d}t} = -\frac{i}{\hbar}\tilde{H}_IU_I, \tag{A. 10}$$

where we recall that $\tilde{H}_I = U_0^\dagger H_I U_0$ by definition of the interaction picture. This equation is subject to the condition $U_I(t_0,t_0) = \mathbb{I}$. We can now work on the operator $U_I(t+\mathrm{d}t,t_0)$, which can be expressed by a Taylor expansion up to order $\mathrm{d}t$ as

$$U_I(t+\mathrm{d}t,t_0) = U_I(t,t_0) + \frac{\mathrm{d}U_I}{\mathrm{d}t}(t,t_0)\mathrm{d}t + o(\mathrm{d}t). \tag{A. 11}$$

We can then inject Eq. (A. 10) into this expansion to get

$$U_I(t + \mathrm{d}t, t_0) = U_I(t, t_0) - \frac{i}{\hbar}\tilde{H}_I(t, t_0)U_I(t, t_0)\mathrm{d}t + o(\mathrm{d}t). \tag{A. 12}$$

By setting $t = t_0$ and using the condition on unitary operator $U_I(t_0, t_0)$ we obtain

$$U_I(t + \mathrm{d}t, t) = \mathbb{I} - \frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t + o(\mathrm{d}t). \tag{A. 13}$$

Note that technically, $\tilde{H}_I(t_0, t_0) = H_I(t_0)$ for the same reason because of U_0. However we are interested into using the expression of $\tilde{H}_I$ in the calculation of Section I.3 and so, we decide to keep the expression in the interaction picture. We can observe that this expansion corresponds to the expansion of the exponential at the same order, indeed

$$e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t} = \mathbb{I} - \frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t + o(\mathrm{d}t), \tag{A. 14}$$

thus proving the expression we used:

$$U_I(t + \mathrm{d}t, t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t}. \tag{A. 15}$$

Appendix B

Some stochastic differential integrations

We want here to justify the expression of Eq. (II.12) where we discarded terms proportional to $\mathrm{d}N_t\mathrm{d}t$. We recall that $\mathrm{d}N_t$ is a Poisson increment, meaning that $\mathrm{d}N_t^2 = \mathrm{d}N_t$. Moreover, we work in a case where the only values this increment can have is 0 or 1, with probability $\mathbb{P}[\mathrm{d}N_t = 0] = 1 - \lambda_t \mathrm{d}t \approx 1$ and $\mathbb{P}[\mathrm{d}N_t = 1] = \lambda_t \mathrm{d}t \approx 0$, where $\lambda_t \in \mathbb{R}$ such that $\lambda_t \gg \mathrm{d}t$.

When dealing with stochastic master equations, it is useful to integrate them in order to understand the underlying mathematical background. Note that the real interest of the SME is to integrate it in order to find the expression of a state. We consider the case of a stochastic differential equation, such as

$$\mathrm{d}f = a\ \mathrm{d}t + b\ \mathrm{d}N_t + c\ \mathrm{d}N_t\mathrm{d}t, \tag{B. 1}$$

where $a, b, c \in \mathbb{R}$, they could be functions of time but we prefer to simplify the discussion. We are going to integrate this equation from t_0 to T, and to do so we discretize the time as $t_0 < t_1 < \cdots < t_n = T$. We thus express the i-th time as $t_i = t_0 + i\delta t$. In particular when $i = n$, we have that $T = t_0 + n\delta t \Leftrightarrow n = \frac{T - t_0}{\delta t}$. Therefore, when $n \to +\infty$, we automatically have $\delta t \to 0^+$. This discretization allows us to define the first integral with measure $\mathrm{d}t$ as

$$\int_{t_0}^{T} a\ \mathrm{d}t = a \lim_{n \to +\infty} \sum_{i=0}^{n-1} (t_{i+1} - t_i) = a \lim_{n \to +\infty} (t_n - t_0) = a(T - t_0), \tag{B. 2}$$

which is simply a Riemann integral. The reason we find a finite quantity is hidden in the fact that we sum infinitely δt while δt is going to 0 at the same time. If we now consider the second integral, that we interpret as a Stieltjes integral, we obtain

$$\int_{t_0}^{T} b\ \mathrm{d}N_t = b \lim_{n \to +\infty} \sum_{i=0}^{n-1} [N(t_{i+1}, t_0) - N(t_i, t_0)] = b \sum_{j=0}^{k-1} 1 = bk, \tag{B. 3}$$

where $N(t_{i+1}, t_0) - N(t_i, t_0) = 0$ most of the time and 1 only k times, according to the probability law followed by the Poisson increment. This result, when applied to the third integral, gives

$$\int_{t_0}^{T} c\ \mathrm{d}N_t\mathrm{d}t = c \sum_{j=0}^{k-1} \mathrm{d}t = ck\mathrm{d}t \approx 0, \tag{B. 4}$$

in the Stieltjes definition of the integral, which explains why we can neglect $\mathrm{d}N_t\mathrm{d}t$ in the photo-detection SME.

Appendix C

Homodyne detection in quantum optics

As explained in Section II.4, the homodyne detection creates a continuous signal containing information on the quadrature of the physical system of interest. We use the notations of Figure 3, i.e., the system field is labelled by the system annihilation operator $\hat{a}$ (and is now an input of the homodyne detection process). We also use another input signal which is a coherent field labelled by its annihilation operator $\hat{b}$. Moreover, we use a 50:50 beam-splitter to form two output fields labelled $\hat{c}$ and $\hat{d}$ which are then measured by two photo-detectors, as depicted in the aforementioned figure. The quadrature of the system is defined by

$$\hat{a}_\theta = \frac{\hat{a}^\dagger e^{i\theta} + \hat{a}e^{-i\theta}}{\sqrt{2}}. \tag{C. 1}$$

Quantum optics [42] assures that, after the beam-splitter, the output fields are

$$\hat{c} = \frac{\hat{a} + i\hat{b}}{\sqrt{2}}, \qquad \hat{d} = \frac{\hat{b} + i\hat{a}}{\sqrt{2}}. \tag{C. 2}$$

These relations hold when the beam-splitter is 50:50. Hence, this exact process is called "balanced" homodyne detection. We explained in Section II.3 that the photo-detector signal is proportional to the mean value of the occupation number. Thus, $I_c \propto \left\langle \hat{c}^\dagger \hat{c} \right\rangle$ and $I_d \propto \left\langle \hat{d}^\dagger \hat{d} \right\rangle$. Therefore, the difference of photo-currents is

$$I_d - I_c \propto i \left\langle \hat{a}^\dagger \otimes \hat{b} - \hat{a} \otimes \hat{b}^\dagger \right\rangle. \tag{C. 3}$$

As explained, the field input labelled by $\hat{b}$ is a coherent field. We denote this field as $|\beta\rangle$ so that $\hat{b}\,|\beta\rangle = \beta\,|\beta\rangle$, where $\beta = |\beta|e^{i\phi} \in \mathbb{C}$. If we denote the input state of the system as $|\psi\rangle$, then the total input state is $|\psi_{\text{in}}\rangle = |\psi\rangle \otimes |\beta\rangle$. We can therefore explicit the average in Eq. (C. 3) as

$$I_d - I_c \propto i|\beta| \left\langle \psi \right| (\hat{a}^\dagger e^{i\phi} - \hat{a}e^{-i\phi}) \left| \psi \right\rangle = |\beta| \left\langle \psi \right| (\hat{a}^\dagger e^{i\theta} + \hat{a}e^{-i\theta}) \left| \psi \right\rangle, \tag{C. 4}$$

where we introduced $\theta = \phi - \pi$. Therefore, we have briefly shown that homodyne detection signal is proportional to the quadrature of the system $\hat{a}_\theta$.

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